

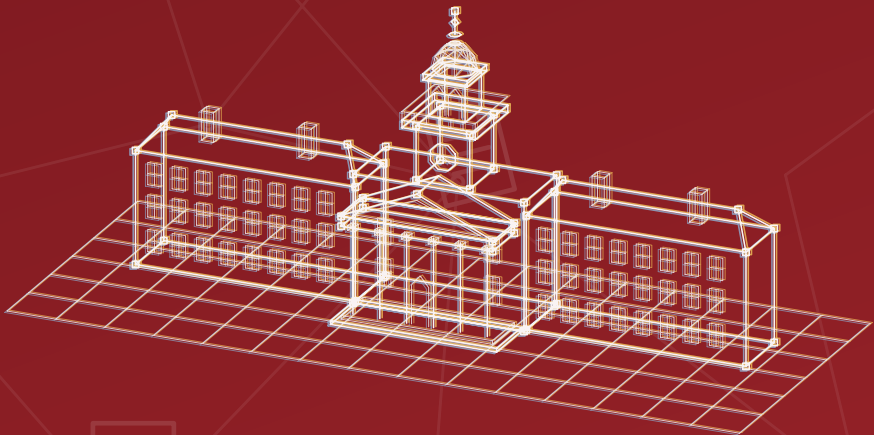


Est. 2020 as a terminal script for ResNet Support Representatives.
Rebuilt 2026 as a monitoring and analytics platform.

BSU Student Printing Ops

The print-station monitor, documented end to end

How Bridgewater State's 31 Wepa print stations are watched, measured, explained and improved, told through the full CRISP-DM lifecycle: from the business problem to the deployed product, with a guide for everyone who uses it.



31 stations · 16 ResNet (green) · 14 Student Computer Labs (gold) · 1 Satellite Campus (white)

DOCUMENT
SPO-DOC-001

PRODUCT VERSION
1.7.0 · October 2026

METHOD
CRISP-DM, six phases

STATUS
Released · living document

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An independent portfolio project. It is not an official publication of Bridgewater State University or Wepa, and it does not represent either. Wepa's status page is read for internal operations only; its data is not republished.

About this document

One document for everyone who touches the product: the student whose printer jammed at 11 PM, the student worker on shift, the supervisor, IT leadership, and the engineer or data scientist who will maintain it next.

Document control	
FIELD	VALUE
Title	BSU Student Printing Ops: product documentation (CRISP-DM lifecycle edition)
Identifier	SPO-DOC-001 , kept with the code in docs/product/
Describes	Product version 1.7.0 (see the change log in the app and in CHANGELOG.md)
Author	Nicholas Freberg
Classification	Internal operations. Contains no personal data; screenshots use the synthetic demo data set.
Dates	Written ISO 8601 (YYYY-MM-DD); times are Eastern (America/New_York), as in the app
Review	Revise with every minor release; the change log names what changed

How to read it

The chapters follow CRISP-DM, the Cross-Industry Standard Process for Data Mining, because the product was built that way: understand the problem, understand the data, prepare it, model it, evaluate the models honestly, then deploy and keep watching. Chapters 7 to 9 step outside that cycle for process models, the user guide and standards.

Every chapter opens with **In short**, a plain-language summary anyone can read. Boxes marked with an audience give a little extra for that reader. Badges at the top of each chapter say who it is mainly for.

Students

Read 8.1 Finding a printer that works. Two minutes.

Student staff

The summary, then the user guide, especially Overview, Rounds and Investigations.

Supervisors

Business understanding, the user guide, and process models.

IT leadership

The summary, evaluation and deployment, then the standards.

Engineers

Data preparation, the data dictionary, architecture and the appendices.

Data scientists

Data understanding through evaluation; the model inventory is in chapter 4.

Contents

0 Executive summary

What it is, why it exists, what it does

1 Business understanding

Problem, stakeholders, goals, journeys, requirements

2 Data understanding

Sources, cadence, quirks, quality

3 Data preparation

Pipeline, cleaning rules, incidents, data dictionary

4 Modeling

Metrics, statistics, machine learning, AI analyst, decision tables

5 Evaluation

Gates, testing, model scoring, limits

6 Deployment

Architecture, Azure, availability, security, updates

7 Process models

BPMN, CMMN, DMN and Six Sigma views

8 User guide

Every page, task by task, for every role

9 Standards and recommendations

ISO, ACM, ASA, OMG, Six Sigma, DASCA, data ecology

A Appendices

Configuration, commands, glossary, sources, credits

A minute-by-minute record of every campus printer

Everyone

IT leadership

Wepa publishes a status page for BSU's print stations that refreshes every minute and only ever shows *now*. BSU Student Printing Ops reads it once a minute, keeps every reading, and turns the history into answers: what is broken, for how long, why, when nobody was there to fix it, what to restock, and where a student should walk instead.

Mission. Keep every BSU print station ready when students need it, by measuring service the way students experience it and acting on what the data shows.

MEASURE	ACT	IMPROVE
Every station, every minute: availability, outages, causes, supplies and usage, computed the same way every time.	Alerts, prioritized rounds, supply forecasts and inventory turn findings into the next visit, the next order and the next fix.	Investigations, evidence for Wepa, placement reviews and reporting close the loop, so service gets measurably better each term.

WHAT IT WATCHES

31 stations: 16 ResNet printers in and around the residence halls, 14 in Student Computer Labs, 1 at a satellite campus.

HOW OFTEN

One reading every 60 seconds, every reading kept. Failed reads are kept too, so gaps are visible rather than guessed.

WHO IT SERVES

Students, ResNet and IT Service Center staff, supervisors and IT leadership, each with their own view.

HOW IT IS BUILT

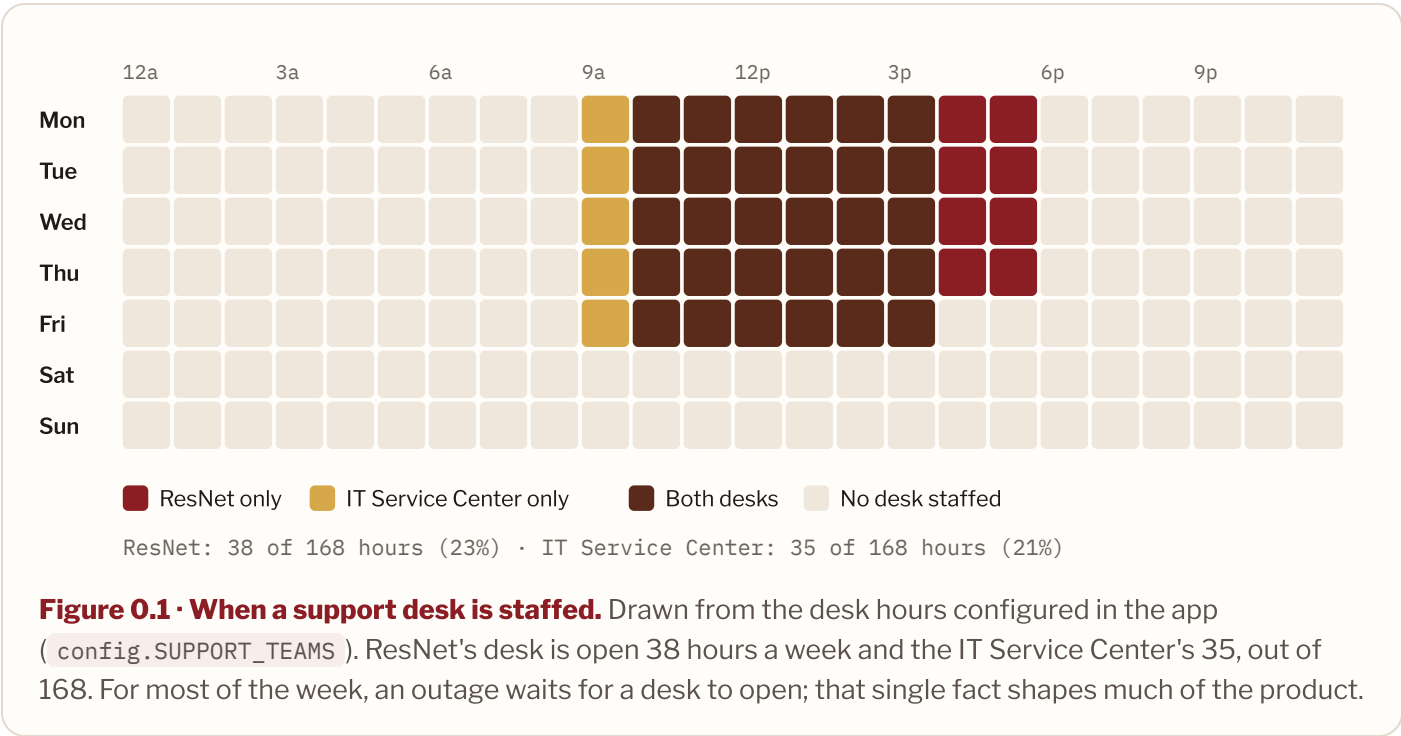
Python, pandas and Dash; about 24,000 lines of application code and more than 200 automated tests; deployed on Azure Container Apps.

What it does

- **Now:** every station's status in one vocabulary (Operational, Degraded, Out of service, No signal), a ranked work queue, and a walking or van route to everything that needs a visit.
- **Then:** availability, time to repair, time between failures, faults by cause and hour, supplies and parts, all with stated definitions and minimum-evidence rules.
- **Why:** plain-language stories, click-to-explain charts, statistical tests, and an optional AI analyst that may only state figures it looked up.
- **Next:** supply reorder points, end-of-life forecasts for parts, and an outage-risk model that is only shown when it beats a simple baseline.
- **Accountability:** permanent reference numbers for every outage, governed investigations with a tamper-evident trail for Wepa cases, named staff accounts and an audit log.
- **Supplies and keys:** an inventory of parts and paper from delivery to the printer each unit went into, invoices read automatically and approved by a second person, counts that open an investigation when stock is missing, weekly paper checks, and a log of who holds a Wepa kiosk key.
- **Self-maintenance:** daily vulnerability checks ranked by real-world exploitation, reviewed update packages, backup collectors and a self-check.

Who uses it, and where to start

AUDIENCE	START HERE	WHAT THEY GET
IT leadership	Executive summary · IT Outcomes	Service levels for the month and year, what changed, and a print-ready annual-report feature
Managers and supervisors	Overview · Analytics · Investigations · Inventory	What needs attention, the station report card, root-cause work with an audit trail, and supply accountability
Data engineers and statisticians	Analytics › Forecasts & statistics · System health	Stated definitions, sample sizes, confidence intervals, model back-tests and data-quality checks
Student workers	Overview · Rounds · Inventory › Record	The next printer to visit, the route, and a place to record deliveries, moves and counts
Students	The student status page	Which printers work right now, and the nearest working one they can walk into



Business understanding

Supervisors

IT leadership

Everyone

IN SHORT

Students need to print, often late and close to a deadline. When a printer breaks, the student finds out at the printer and staff find out later. The product exists to shorten that gap, to show where the gaps come from, and to give the university evidence to act on, without adding work for student staff.

1.1 Background

Bridgewater State University offers Wepa print stations across campus: in residence-hall areas (supported by ResNet, the residential network team based in East Campus Commons), and in labs, the library, the student union and a satellite campus (supported by the IT Service Center in Maxwell Library). Students pay per page through Wepa and print papers, forms, tickets and boarding passes.

Wepa publishes a public status page listing every station with a color (green, yellow, red), alert codes, printer text and consumable levels. It refreshes every minute and shows only the present. The original version of this product (2020) was a terminal script for ResNet Support Representatives: choose your assigned area and it beeped when a printer there went down.

1.2 Problem statement

Nobody could answer basic operational questions with evidence: which printers break most; how long outages last and why; whether problems wait overnight because the desks are closed; what supplies will run out next; and where a student should go when their printer is down. Without history, recurring hardware faults looked like bad luck, and conversations with the vendor relied on memory.

1.3 Stakeholders

STAKEHOLDER	NEEDS	USES
Students	A working printer nearby, especially after hours	Student status pages (QR signs), indirectly everything
ResNet Support Representatives	Know what is broken, where to go first, what to carry	Overview, Rounds, station pages, Investigations
IT Service Center staff	The same, for labs and public spaces	Overview, Rounds, Stations
Supervisors (ResNet, ITSC)	Coverage, staffing, recurring faults, vendor evidence	Analytics, Planning, Investigations, Insights
IT leadership	Service level, trends, outcomes to report	Executive summary, IT Outcomes, exports
Residence Life	Printing in the halls students live in	Reports from supervisors
Wepa (vendor)	Clear, dated evidence when hardware fails repeatedly	Investigation evidence packets (PDF)

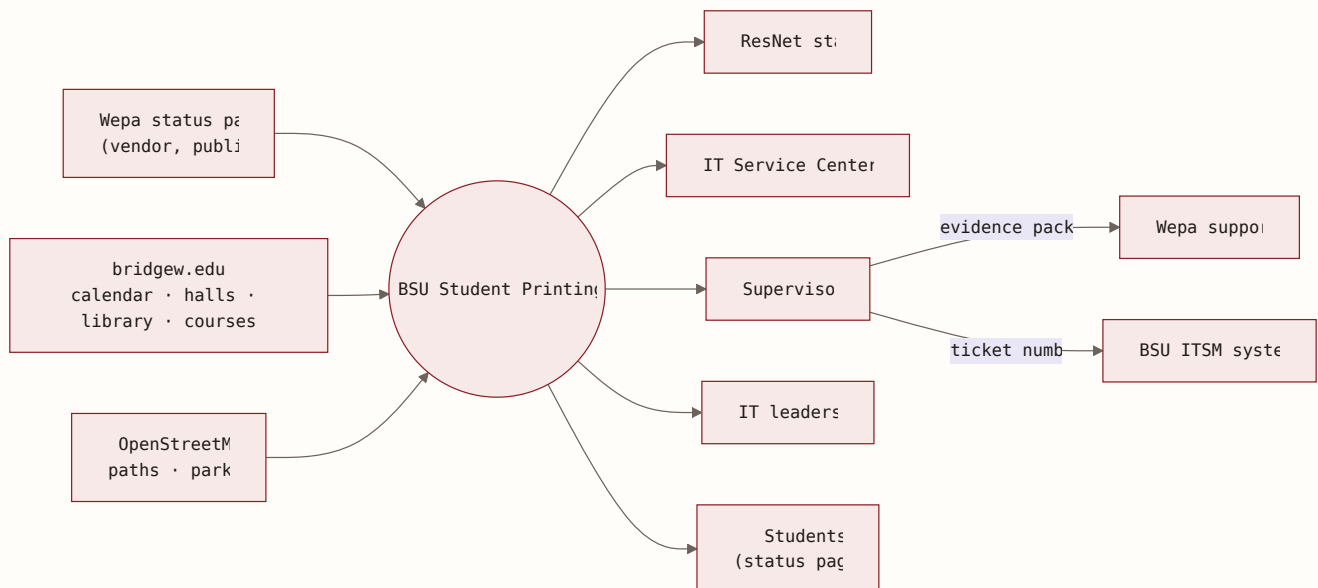


Figure 1.1 • Context. What flows into the product and who it serves. The ITSM system and Wepa support are referenced by number only; nothing is synchronized with them.

RESPONSIBILITIES (PROPOSED RACI)

ACTIVITY	STUDENT STAFF	SUPERVISOR	IT LEADERSHIP	ADMINISTRATOR	WEPA
Respond to an outage	R	A	I	–	C
Open and work an investigation	R	A	I	–	C
Review and close an investigation	–	R/A	I	–	–
Escalate a hardware pattern to Wepa	C	R/A	I	–	R (fix)
Monthly and annual reporting	–	R	A	–	–
Accounts, deploys, security updates	–	C	I	R/A	–

R responsible · A accountable · C consulted · I informed. This is a recommendation for how the roles map onto the product; it is not a published BSU policy.

1.4 Objectives and success criteria

CRISP-DM separates business objectives from the data-mining goals that serve them. Success is measured by the product's own figures once enough live history exists.

BUSINESS OBJECTIVE	DATA GOAL	SUCCESS CRITERION
Fix printers sooner	Detect each outage within a minute; rank the queue by impact	Median time to repair falls term over term
Fewer outages that wait overnight	Split downtime into desk hours and after hours per team	Share of downtime that starts after hours is known and tracked
Evidence for the vendor	Permanent reference numbers; recurring-fault detector; hash-chained investigations	Every Wepa case backed by a dated evidence packet
Supplies before they run out	Usage that survives part replacements; end-of-life forecasts; reorder points	No station out of a consumable that was forecast
Point students to a working printer	Walking-time network between buildings; access rules	Every printer has a named backup and walking time
Trustworthy numbers	Data-quality score; evidence gates; no imputation	Data quality 95 or above; every figure states its basis

1.5 SIPOC

A Six Sigma SIPOC places the product in the service it supports: suppliers, inputs, process, outputs, customers.

SUPPLIERS	INPUTS	PROCESS	OUTPUTS	CUSTOMERS
Wepa BSU web pages OpenStreetMap Open-Meteo Vulnerability databases	Station status every minute Calendar, halls, library hours, courses Paths, parking Hourly weather Advisories	Collect → store → detect incidents → measure → explain → act → review	Live status and work queue Routes KPIs and reports Investigations and evidence Alerts	Students Student staff Supervisors IT leadership Wepa

1.6 Voice of the customer, critical to quality

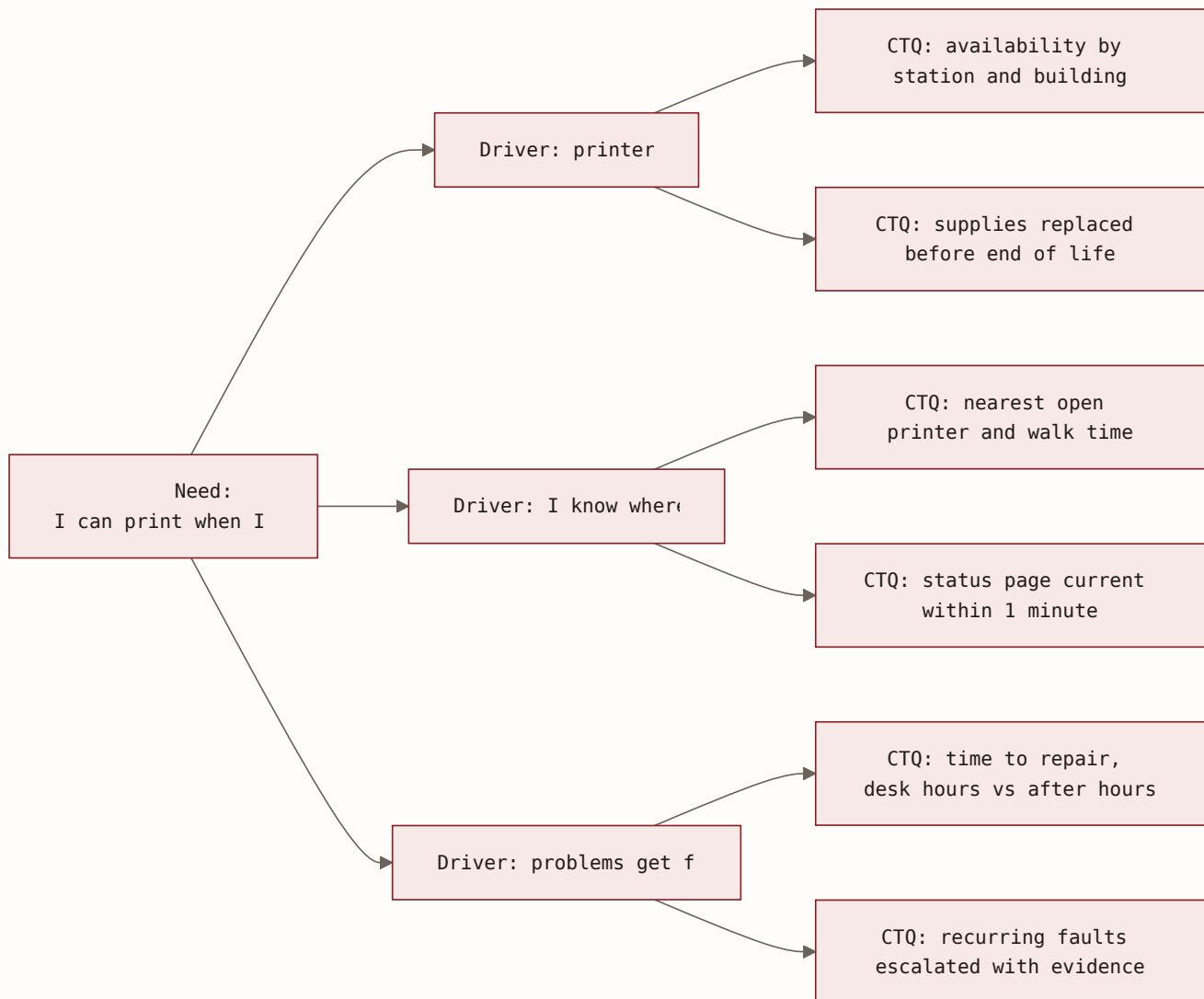


Figure 1.2 · CTQ tree. From the student's need to measurable characteristics the product reports.

1.7 Customer journeys

Three journeys, scored 1 (frustrating) to 5 (smooth). The scores are design assessments of the experience with the product in place, not survey results.

A student, 11 PM, paper due at 8

- ResNet
- Student

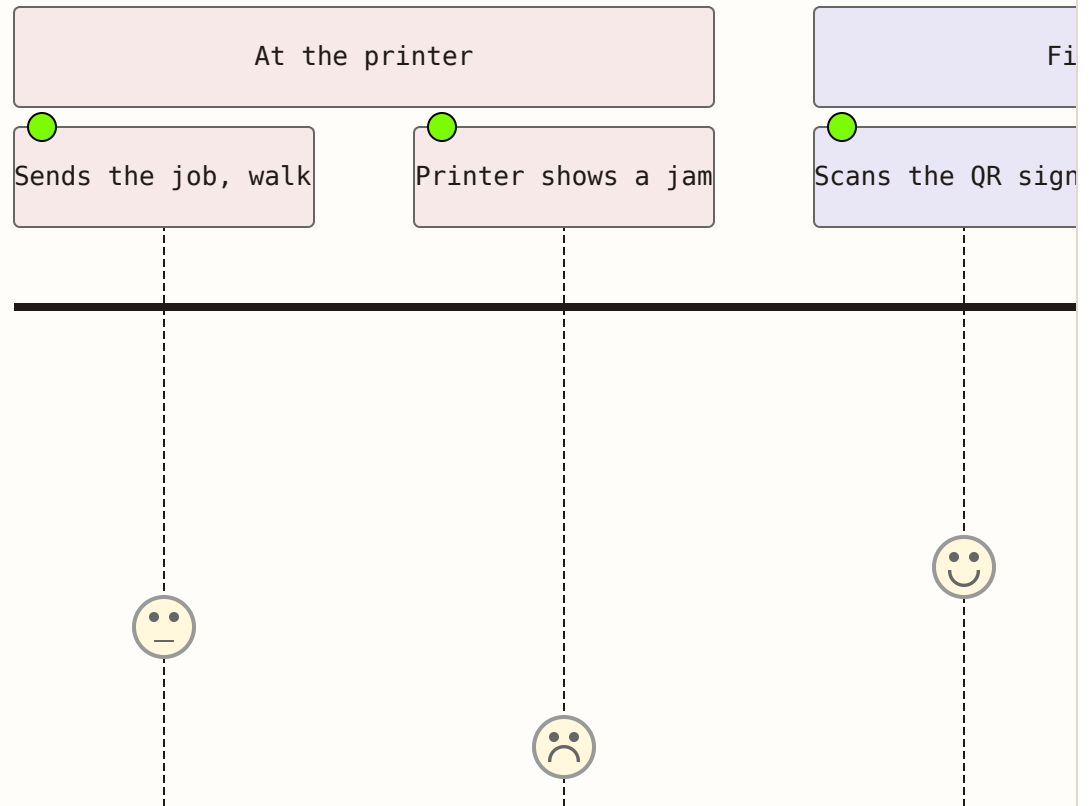


Figure 1.3 • Student journey. The QR sign and the "nearest open printer" rule turn a dead end into a short walk. The outage was recorded at the minute it started, so staff do not depend on a report.

A ResNet Support Representative

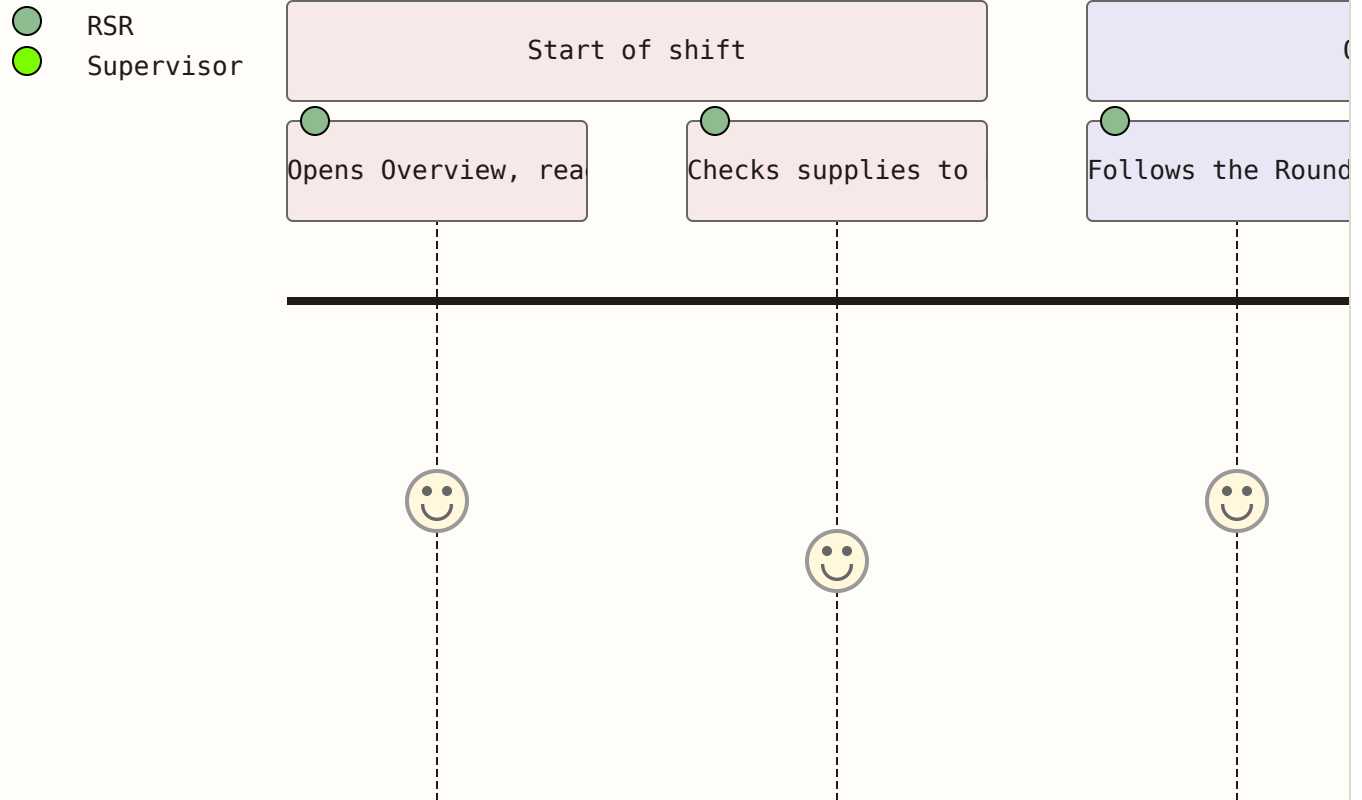


Figure 1.4 · Staff journey. The detector opens an investigation when one printer has three or more unreachable or hardware outages in 14 days.

IT leadership at month end

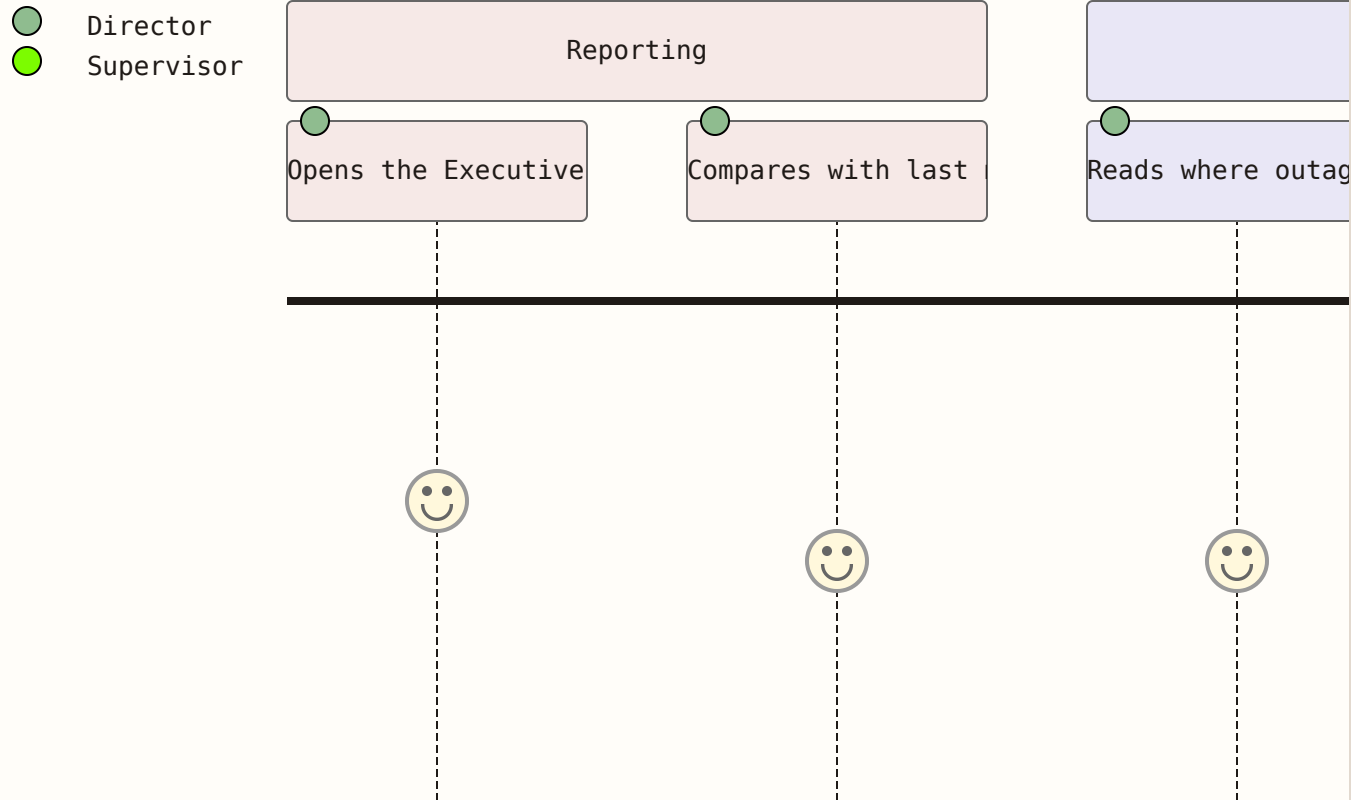


Figure 1.5 · Leadership journey. The same numbers staff work from, summarized, with their definitions attached.

1.8 Constraints, assumptions and risks

TYPE	ITEM	RESPONSE
Constraint	Wepa's terms allow personal, non-commercial use of its site	Data is used for internal operations; not republished; written permission from Wepa and BSU before any public use
Constraint	The status page can change format without notice	Columns are found by header text; a mismatch fails loudly and is logged
Constraint	No access to Wepa's internal systems or job logs	Usage is inferred from consumable levels; stated as such
Assumption	The row color is Wepa's own verdict of severity	Used directly; codes only add names and fix categories
Assumption	Desk hours in configuration are the dependable coverage	Editable in one place; holidays from the academic calendar
Risk	Monitoring gaps (network, restarts)	Gaps recorded and excluded, never imputed; backup collectors
Risk	Over-reading thin evidence	Minimum-evidence gates; sample sizes on every statistic
Risk	AI stating something untrue	Read-only tools, a fact check of every number, links and HTML stripped
Risk	Vulnerable dependencies	Hash-pinned installs, daily scans, threat-intelligence ranking

1.9 Requirements

Requirements are prioritized with MoSCoW and traced to the module that implements them. Status reflects version 1.7.0. Quality requirements are grouped by the ISO/IEC 25010 characteristic they serve.

Functional requirements

ID	REQUIREMENT	PRIORITY	STATUS	WHERE
FR-01	Read every station's status once a minute and log every attempt, successful or not	Must	Done	scrape.py , collector.py
FR-02	Keep raw readings append-only; never edit them	Must	Done	store.py
FR-03	Turn readings into outages, warnings, fault and paper-tray incidents with start, end and duration	Must	Done	events.py
FR-04	Count consumable usage correctly across part replacements	Must	Done	consumables.py
FR-05	Show every station's current status in one vocabulary	Must	Done	vocab.py , Overview
FR-06	Publish KPIs with written definitions and minimum-evidence gates	Must	Done	metrics.py
FR-07	Give every outage a permanent reference number	Must	Done	refs.py
FR-08	Named staff accounts managed centrally; sign-in on every page	Must	Done	accounts.py , auth.py
FR-09	Monitor data quality and the collector itself	Must	Done	System page
FR-10	Clearly labelled synthetic demo mode	Must	Done	synth.py
FR-11	Ranked work queue and a route planner (walking or van)	Should	Done	ops.py , routing.py
FR-12	Nearest printer open to everyone, with walking time	Should	Done	nearby.py
FR-13	Split downtime by desk hours; say when support is unavailable	Should	Done	support.py
FR-14	Analytics: reliability, faults, supplies, usage, report card, planning, statistics	Should	Done	Analytics page

FR-15	Plain-language stories, Ask the data, click-to-explain	Should	Done	narrative.py , ask.py
FR-16	Governed investigations with a tamper-evident history and evidence packet	Should	Done	investigations.py
FR-17	Activity feed and notifications	Should	Done	activity.py
FR-18	Export any view as CSV, Excel, JSON or PDF	Should	Done	export.py
FR-19	Campus context: calendar, halls, library hours, class schedules	Should	Done	campus.py , courses.py
FR-20	Daily vulnerability check, ranked by exploitation	Should	Done	vulns.py , threatintel.py
FR-21	Self-check and OWASP Top 10 checklist	Should	Done	selfcheck.py , owasp.py
FR-22	Executive summary and IT Outcomes report page	Could	Done	insights.py
FR-23	Outage-risk model, shown only when it beats a baseline	Could	Done	risk.py
FR-24	AI analyst with read-only tools and a fact check (optional)	Could	Done	ai.py , analyst.py
FR-25	Student status pages and printable QR signs (opt-in public)	Could	Done	dashboard/public.py
FR-26	Update packages built and tested by a reviewed pipeline	Could	Done	updates.py
FR-27	Backup collectors with a signed handshake	Could	Done	peer.py
FR-28	Feature requests filed as GitHub issues	Could	Done	feedback.py
FR-29	Campus map and Google Earth export	Could	Done	geo.py
FR-31	Inventory of parts and paper by location, with a trail of every unit, second-person approval of deliveries and investigations for shortfalls	Should	Done	inventory.py , ledger.py

FR-32	Read supply invoices (PDF or photo) into draft lines for approval	Could	Done	inventory.py , ai.py
FR-33	Wepa kiosk key log; a lost key opens an investigation (staff only, never shared with AI)	Should	Done	keys.py
FR-34	Grouped navigation, one filter bar and a phone tab bar	Should	Done	dashboard/app.py
FR-30	Single sign-on with BSU's Entra ID (MFA)	Should	Planned	Needs BSU IT registration

Quality requirements (ISO/IEC 25010 characteristics)

ID	CHARACTERISTIC	REQUIREMENT	HOW IT IS MET
NFR-01	Reliability	The collector never stops on an error	Failures recorded as data; the loop continues
NFR-02	Reliability (availability)	No collection gap from a deploy or a collector failure	Rolling deploys with a collector lock; backup takes over after 60 s of silence and stands down within one 15 s check
NFR-03	Performance efficiency	Pages reflect a new reading within about 15 s	Incremental refresh; finished days read once (Parquet)
NFR-04	Security (confidentiality)	Every route needs a session; least privilege	Tested by walking every route; roles; CSP
NFR-05	Security (integrity)	Records provably unaltered	Append-only files; SHA-256 hash chain; hash-pinned installs
NFR-06	Interaction capability (usability)	Readable by anyone	One vocabulary; whole-unit durations; plain-language stories
NFR-07	Accessibility	Keyboard, contrast and motion preferences respected	Appearance menu; visible focus; reduced motion. A formal WCAG 2.2 audit is recommended
NFR-08	Maintainability	Rules changeable in one place; changes verified	<code>config.py</code> ; 190 automated tests; lint and scans in CI
NFR-09	Flexibility (portability)	Runs on a laptop, a VM or containers	Python 3.11/3.12; three documented deployments
NFR-10	Privacy	Collect the minimum about people	Visits shortened to network; no passwords logged; names kept out of GitHub issues
NFR-11	Statistical honesty	Never present a guess as a measurement	No imputation; evidence gates; sample sizes; demo labelled
NFR-12	Cost control	Optional AI cannot run away with spend	Calls per hour capped; answers cached

Data understanding

Data scientists

Engineers

Supervisors

IN SHORT

One web page is the heart of it: Wepa's status page, read every minute. Around it sit campus facts (the academic calendar, residence halls, library hours, class schedules), campus maps for walking routes, hourly weather, and security advisories. Each source has known quirks, and each is handled openly rather than smoothed over.

2.1 Sources

SOURCE	WHAT IT PROVIDES	CADENCE	TERMS AND NOTES
Wepa status page <code>cs.wepanow.com/000BRIDGEW149.html</code>	Per station: color (green, yellow, red), alert codes, printer text, toner, drum, belt and fuser levels, page timestamp	Every 60 s	Personal, non-commercial use under Wepa's terms; internal operations only
bridgew.edu	Academic calendar, residence halls, library hours	Calendar yearly, library weekly	Public university pages; committed copies kept as fallback
BSU course search	Class meetings by building and room	Monthly	Public schedule search
OpenStreetMap	Footpaths, roads and parking for routes and walking times	On demand	© OpenStreetMap contributors, ODbL
Open-Meteo	Hourly Bridgewater weather (humidity)	Daily	CC BY 4.0 data
OSV.dev, CISA, NIST NVD, Microsoft MSRC, FIRST EPSS, Exploit-DB, Metasploit	Known vulnerabilities in the app's own packages, and how urgent they are	Daily (catalogs weekly)	Only package names, versions and CVE numbers leave the server
Reference files (maintained by hand)	Station to building, area, type and access; building coordinates; van parking	When the campus changes	<code>reference/stations.csv</code> , <code>buildings.csv</code> , <code>parking.csv</code>

2.2 What a reading looks like

Each minute produces one row per station. A trimmed example, as stored:

SCRAPE_TS	SECTION	STATION_ID	DESCRIPTION	ROW_STATUS	STATUS_CODE
2026-10-02T23:41:00Z	ResNet	02132	East Campus Commons	red	paper_out_er

Illustrative row in the stored format. Levels are percentages remaining; codes are normalized (`Alert_paper_out_error` becomes `paper_out_error`).

2.3 Known quirks

- **Severity comes from the row color.** Wepa's own green, yellow or red verdict is used directly. Codes add a name and a fix category; only a row with no color falls back to the code table.
- **Codes carry details.** `Alert_Paper_Low_Letter_530_sheets_left` is folded into `paper_low` so counts don't split by sheet count.
- **Only some codes have been seen live.** Three red codes and one yellow code are confirmed on BSU's page; the rest come from the page's legend and are marked "inferred" until they appear.
- **Levels jitter.** Sensors wobble a point or two (51, 50, 51, 50); a replacement jumps by 15 or more points.
- **The page can change.** Columns are matched by header text. A rename or reorder fails the read loudly and is logged, instead of writing values into the wrong fields.
- **Reads fail sometimes.** Network blips, Wepa maintenance and restarts all happen. Every attempt is logged, so a gap is a measured fact.

2.4 Status codes

CODE	NAME	FALLBACK SEVERITY	FIX CATEGORY	SEEN ON BSU'S PA
paper_out_error	Paper out	red	Refill / reseal paper	Yes
printer_down	Printer down	red	Inspect printer	Yes
tray_missing	Tray missing	red	Refill / reseal paper	Yes
paper_low	Paper low	yellow	Refill / reseal paper	Yes
paper_jam	Paper jam	red	Clear jam	Inferred
not_reachable	Not reachable	red	Check network / restart station	Inferred
fatal_error , service_call	Fatal error, service call	red	Power-cycle printer	Inferred
cover_open	Top cover open	red	Inspect printer	Inferred
toner_critical , drum_critical , toner_sensor_error	Consumable critical	red	Replace consumable	Inferred
toner_low , incorrect_tray_size	Toner low, wrong tray size	yellow	Replace consumable / paper	Inferred

2.5 Data quality

Data quality is measured, scored and shown on the System page, so every figure can be read with its reliability in mind.



Figure 2.1 · How the data quality score is built. The weights are fixed in `metrics.data_quality`: freshness (time since the last good reading), completeness (readings received versus one per station per minute), validity (levels within 0 to 100) and station coverage.

FOR DATA SCIENTISTS

Observed time is explicit. Each reading covers the time until the station's next reading, up to 5 minutes; anything longer is unobserved and excluded from every rate. Availability is therefore a ratio of observed minutes, never of wall-clock minutes.

FOR SUPERVISORS

A low data-quality day means that day's numbers rest on less evidence, not that they are wrong. The System page shows which days, and the live log shows why.

Data preparation

Engineers

Data scientists

IN SHORT

Raw readings are stored exactly as read and never edited. Everything else is computed from them: when each outage started and ended, how much toner was really used, how long printers were available. If a rule changes, the whole history is simply recomputed.

3.1 Pipeline

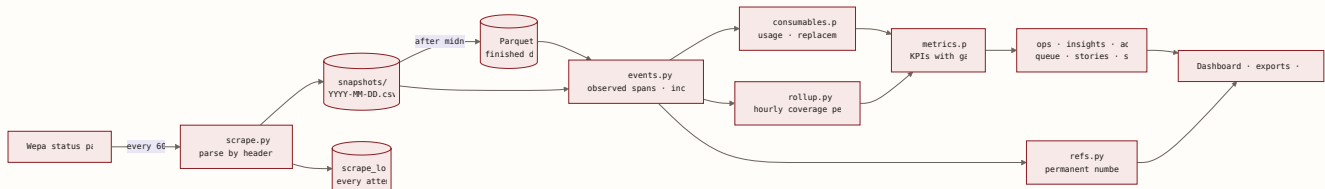


Figure 3.1 · From page to dashboard. Storage is append-only CSV for today and Parquet for finished days. Imported history (from another computer or a backup collector) sits beside it and is merged on load, with duplicate minutes dropped.

3.2 Cleaning and construction rules

One reading per minute

When two collectors overlap (a deploy, a backup), duplicates for the same station and minute are dropped; for the scrape log, a success wins over a failure.

Observed time

A reading covers the time to the next one, up to 5 minutes. Longer gaps are unobserved: never counted as up or down.

Incidents

Consecutive readings of one station in the same state. An incident opens at the first reading showing it and resolves at the first that doesn't. Gaps under 6 hours with the same state on both sides are bridged.

Censored starts

An incident already open when monitoring began has an unknown start. It is kept (it is real) but excluded from repair-time averages.

Replacements

A rise of 15 points or more is a part replacement, logged with what the old part had left, and never counted as negative usage.

Usage

Between replacements, usage is the first reading minus the lowest reading so far, so jitter is counted once. 100 points equals one part's worth.

Desk hours

Each outage is split into time while its team's desk was staffed and time after hours, using the configured hours and the academic calendar's holidays.

Reference numbers

Every outage gets a permanent number by its cause: **OUT** unreachable, **JAM** jammed, **PAP** paper, **SUP** consumable out, **ERR** hardware or system; investigations **INV**, update packages **UPD**, feature requests **REQ**. Nine digits; never reused.

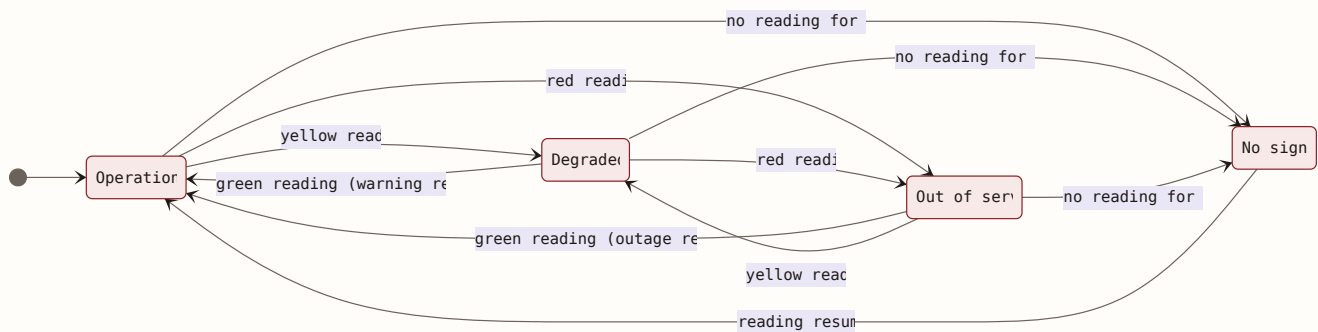


Figure 3.2 • Station states. The four words used everywhere. "No signal" is not a fault; it means the station's status is unknown, and that time is left out of every rate.

3.3 Worked example: the replacement rule

TIME	BLACK TONER	COUNTED AS	RUNNING TOTAL
09:00	12%	–	0
11:00	7%	usage +5	5
13:00	3%	usage +4	9
13:20	100%	replacement (3% left in the old cartridge)	9
16:00	94%	usage +6	15

3.4 Data dictionary

Every stored field the product reads or writes. Times are UTC in storage and shown in Eastern time. Records under `records/` are append-only JSON lines unless noted.

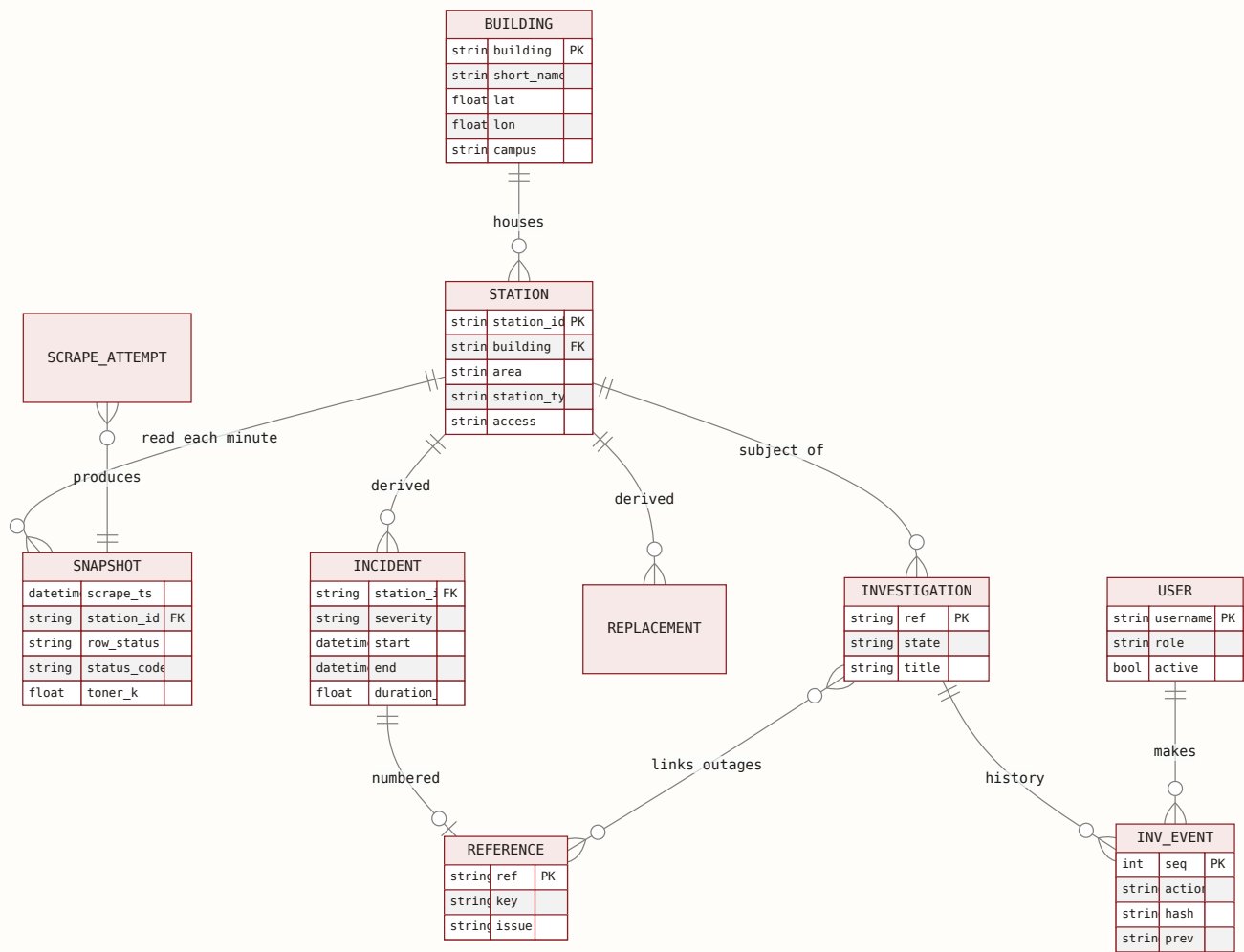


Figure 3.3 · Entities and relationships. Stored entities (snapshots, attempts, references, investigation events, users) and derived ones (incidents, replacements) computed on load.

snapshots/ · one row per station per successful read

FIELD	TYPE	MEANING
scrape_ts	datetime, UTC	When the page was read
page_ts	datetime, UTC	The timestamp printed on Wepa's page
section	text	Page section: ResNet, Student Computer Labs, Satellite Campuses
station_id	text, 5 digits	Wepa's station number (P.S.#); key to the reference file
description	text	Wepa's station description
row_status	green / yellow / red	Wepa's verdict for the row
status_codes	text, ; - separated	Normalized alert codes
printer_text	text	Printer message, e.g. a tray warning
toner_k c m y	number 0–100	Toner remaining, black, cyan, magenta, yellow
drum_k c m y	number 0–100	Drum life remaining
belt , fuser	number 0–100	Transfer belt and fuser life remaining

scrape_log/ · one row per attempt, successful or not

FIELD	TYPE	MEANING
attempt_ts	datetime, UTC	When the read was attempted
ok	boolean	Whether the page was read and parsed
http_status	integer	HTTP status code, if any
duration_ms	integer	How long the read took
n_stations	integer	Stations found on the page
error	text	Why it failed (never contains credentials)

reference/stations.csv · maintained by hand

FIELD	TYPE	MEANING
station_id	text	Wepa station number
building	text	Building name; key to buildings.csv
area	text	Campus area, e.g. University Park
station_type	residence / lab / satellite	What kind of place it serves
access	public / residents	Who can walk up to it. Blank means residents for residence-hall printers and public otherwise. East Campus Commons is public : a ResNet printer in a building every student uses
notes	text	Location detail

Incidents (derived on load) · outages and warnings

FIELD	TYPE	MEANING
station_id	text	Station
severity	red / yellow	Outage (out of service) or warning (degraded)
start , end	datetime	First reading showing it; first reading after it cleared
status	open / resolved	Shown as Ongoing or Resolved
censored_start	boolean	Already open when monitoring began
duration_s	seconds	Length, shown in days, hours and minutes
staffed_s , after_s , wait_s	seconds	Time inside and outside desk hours; wait until the desk opened
ref	text	Permanent reference number (outages only)

records/ · the product's own records

FILE	ONE LINE PER	KEY FIELDS
<code>refs.jsonl</code>	Reference number issued	<code>ref</code> , <code>key</code> (station and start), <code>issue</code> , <code>created</code>
<code>investigations.jsonl</code>	Change to an investigation	<code>seq</code> , <code>ts</code> , <code>inv</code> , <code>user</code> , <code>action</code> , <code>changes</code> (old and new), <code>note</code> , <code>prev</code> , <code>hash</code>
<code>system_events.jsonl</code>	App event for the activity feed	<code>kind</code> , <code>title</code> , <code>detail</code> , <code>href</code>
<code>versions.jsonl</code>	Version first run	<code>version</code> , <code>previous</code> , <code>commit</code>
<code>updates.jsonl</code>	Update package change	<code>ref</code> (UPD), <code>state</code> , <code>spec</code>
<code>feedback.jsonl</code>	Feature request change	<code>ref</code> (REQ), <code>kind</code> , <code>state</code> , <code>issue</code>
<code>vulns.json</code> (snapshot)	Latest vulnerability check	findings with ids, severity, fix, priority, sources
<code>security/users.json</code> (snapshot)	Staff directory	username, name, email, role, active, resident, hall, phone, script hash, session epoch

Modeling

Data scientists

Supervisors

IT leadership

IN SHORT

Most of the product's value comes from careful definitions, not exotic models: what counts as available, how long an outage really lasted, what usage really was. On top of those sit a handful of well-understood statistical methods, each chosen to answer one operational question, and one machine-learning model that has to prove itself before it is shown.

4.1 Measures

MEASURE	DEFINITION	SHOWN WHEN
Availability	Observed printer-minutes not out of service ÷ all observed printer-minutes	At least half the window observed
Building coverage	Share of minutes at least one printer in the building could print	Same
Mean time to repair	Mean duration of resolved outages (or warnings), censored starts excluded	At least 3 incidents
Mean time between failures	Hours of uptime ÷ number of outages	At least one outage
Paper refill time	Mean duration of paper-out outages	At least 3 incidents
Tray empty time	Mean time a single tray reports empty while others may still print	At least 3 incidents
Burn rate	Consumable points used per day, from the last 14 days	At least 3 days observed
Data quality	30% freshness + 40% completeness + 20% validity + 10% station coverage	Always

Below the evidence threshold the dashboard says "insufficient data" rather than showing a number that would swing wildly with one more incident.

4.2 Model inventory

QUESTION	METHOD	WHY THIS METHOD
When will this part reach end of life?	Theil–Sen regression on readings since the last replacement, with a 90% interval	Robust to sensor blips; falls back to burn rate with few readings
Is today's fault count normal?	Statistical process control c-chart (mean $\pm 3\sqrt{\text{mean}}$) and the Western Electric 8-in-a-row rule; Poisson tail test per station-day	Counts of rare events; the standard Six Sigma control tool
How fast are outages fixed, in and out of desk hours?	Kaplan–Meier survival curves with a log-rank test	Keeps still-open outages as censored rather than dropping them
Do busier printers fail more?	Ordinary least squares, outages per week on usage, with residual flags beyond 2σ	Separates "busy" from "bad unit"
Which printers fail most, fairly?	Empirical-Bayes Gamma–Poisson rates with 90% credible intervals; report-card grades shrink toward the campus rate	Small samples don't top the list by chance
Did several printers fail together?	Clusters of outages within 10 minutes across 3+ stations, tested against a Poisson rate for that hour and day type	Tells a network or Wepa-side event from a busy moment
Which downtime cost the most printing?	Outage hours weighted by that printer's typical use at that time of week	An hour down at 2 PM on a Tuesday matters more than at 4 AM
Does humid weather go with more jams?	Mantel–Haenszel rate ratio, stratified by 6-hour block and weekday or weekend	Controls for the fact that humidity and printing both follow the clock
What if a desk stayed open later?	Monte Carlo replay (1,000 runs) of real outages with extra desk coverage; fix times drawn from real desk-hour fixes	Answers staffing questions with the campus's own history
Where would one more printer help most?	Greedy one-step p-median on walking times, weighted by busy-weighted downtime	Directly minimizes students' extra walking
Supplies to reorder	Monte Carlo of recent daily usage: how many of each part to stock at 50%, 90% and 95%	Gives a range, not a single guess

confidence; reorder points use the supply lead time (7 days unless set)

Which printer may fail in the next 24 hours?

Machine learning, below

Only if it beats the simple alternative

4.3 The outage-risk model

Each printer-hour becomes one example: time of day and week, whether a desk is open and how long until it opens, whether classes are in session, the kind of place, recent faults and warnings, recent usage, part levels. The label is whether an outage started in the next 24 hours.

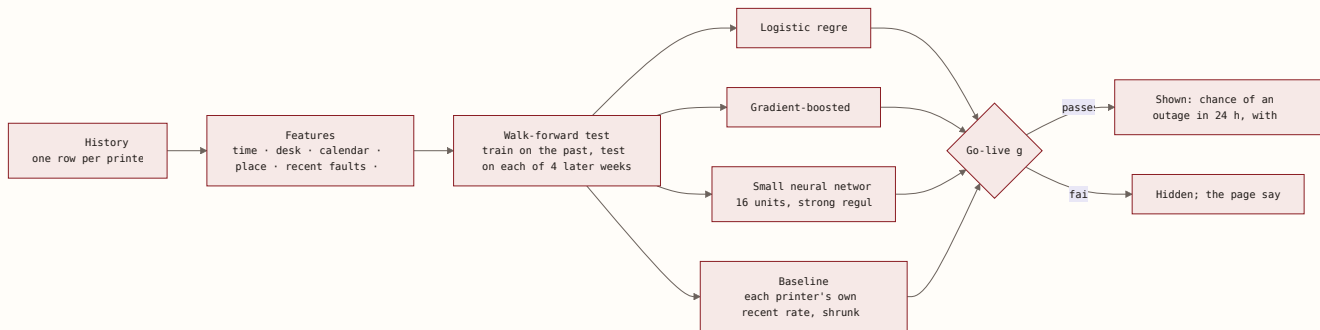


Figure 4.1 · Train, test, gate. Recent history counts more (weight halves every 21 days). The model retrains nightly; every prediction is logged so its track record can be checked later.

4.4 Decision tables (DMN)

The product's key decisions are written as DMN-style decision tables. The hit policy is shown in the top-left cell: **U** unique (exactly one rule applies), **F** first (the first matching rule wins), **A** all conditions must hold.

Decision: station status

F	LAST READING AGE	ROW COLOR	STATUS SHOWN
1	over 5 minutes	–	No signal
2	≤ 5 minutes	red	Out of service
3	≤ 5 minutes	yellow	Degraded
4	≤ 5 minutes	green	Operational

Decision: show the outage-risk model

A	CONDITION	THRESHOLD
1	Days of history	≥ 21
2	Outages in training	≥ 50
3	Outages in the test weeks	≥ 25
4	Ranking skill (AUC) of the best model	≥ 0.65
5	Brier skill over the baseline	≥ 2%

Decision: vulnerability priority

F	EVIDENCE FROM THE SOURCES	PRIORITY
1	On CISA's Known Exploited list, or CISA or Microsoft report active exploitation, or a Metasploit module exists	Act now (Soon for install-only tools)
2	Public exploit on Exploit-DB, CISA proof of concept, automatable, EPSS ≥ 10%, or high or critical severity	Soon
3	None of the above	Routine

Decision: investigation impact

F	HISTORY	SCORE (USAGE + REDUNDANCY + LOST PRINTING, EACH 1–3)	IMPACT
1	under 90 days	–	Moderate (default)
2	90 days or more	average under 1.5	Low
3	90 days or more	average 2.5 or more	High
4	90 days or more	otherwise	Moderate

Decision: backup collector

U	BACKUP STATE	MAIN COLLECTOR	ACTION
1	Standing by	live	Stay on standby; update itself if the main app is newer
2	Standing by	silent under 60 s	Keep checking every 15 s
3	Standing by	silent 60 s or more	Take over: collect every minute, send readings to the dashboard
4	Collecting	live again	Stop at once, report the period covered, send everything collected
5	Collecting	still silent	Keep collecting

4.5 The AI analyst

The Assistant is optional and runs only with an API key. It is an analyst with 16 read-only tools over the same data the pages use (station profiles, availability, incidents, repair times, supplies, usage, the report card, period comparisons, campus context, statistics, the risk model, investigations and the app's self-check).

- **Facts only.** Every number it writes is checked against the tool results it received, allowing only honest unit restatements (minutes to hours, shares to percentages). Unsupported figures trigger one rewrite; anything still unsupported is flagged.

- **Read-only.** It cannot change records, start updates or contact anyone.
- **Safe output.** Answers are shown without links, images or HTML.
- **Scoped and fair.** BSU printing only; never guesses who caused a problem; says "demo data" when it is.

Evaluation

IT leadership

Data scientists

Engineers

IN SHORT

Each output is checked three ways: does the code do what the definition says (automated tests), is the evidence strong enough to show it (gates), and does the model actually beat the obvious alternative (held-out testing). Where the answer is no, the product says so instead of showing a confident number.

5.1 Does it meet the business goals?

OBJECTIVE	EVIDENCE IN THE PRODUCT	ASSESSMENT
Fix printers sooner	Outage detected at the first red reading; ranked queue; routes	Met in design Outcome needs a term of live data
Fewer outages waiting overnight	Desk-hours split per team; staffing what-if	Met
Evidence for the vendor	Reference numbers; detector; hash chain; evidence PDF	Met
Supplies before they run out	Replacement-aware usage; end-of-life and reorder forecasts	Met Accuracy improves with history
Point students to a working printer	Walking network; access rules; QR status pages	Met
Trustworthy numbers	Quality score; gates; no imputation; definitions	Met

5.2 How the outage-risk model is scored

Walk-forward

Train on everything before a week, test on that week, for each of the last four weeks. The model never sees the future it is tested on.

Brier score	Mean squared error of the predicted probabilities; lower is better. Skill is the improvement over the baseline's Brier score.
AUC	How well the model ranks printers that did fail above those that didn't; 0.5 is a coin flip.
Calibration	Predicted chance versus observed rate in bins, shown as a chart, so "20%" means about 1 in 5.
Track record	Live predictions are logged and later compared with what happened.

FOR DATA SCIENTISTS

The baseline is deliberately strong: each printer's recent outage rate, shrunk toward the campus rate. An early version's baseline scored below 0.5 AUC because raw per-printer rates were noisy; shrinkage fixed it. A model that cannot beat this baseline is not shown.

FOR LEADERSHIP

"Not enough evidence yet" is a correct answer. The risk model stays hidden on new deployments until the data supports it; the page explains which condition is not met.

5.3 Software verification

- **190 automated test cases** run on every push and daily: the page parser against saved real pages, incident rules, the replacement rule, desk hours, the calendar, stories, Ask the data, routing (including an exact Held-Karp check), investigations, accounts and sign-in, security probes, the update pipeline, backup collectors, threat intelligence and durations.
- **Lint (ruff), static security analysis (bandit) and known-vulnerability scanning** (pip-audit over the exact lock file) in continuous integration.
- **Attacker-style tests:** every route without a session, path traversal, oversized uploads, cross-site posts, password guessing, forged or replayed backup messages, tampering with the audit trail.

5.4 Known limits

- Usage is inferred from consumable levels; Wepa's job counts are not available.

- Several alert codes are inferred from Wepa's legend until they appear on BSU's page.
- Forecasting across semesters (ARIMA and similar) is deliberately postponed until a full academic year of live history exists.
- Demo data is synthetic. On it, the models mostly rediscover patterns built into the simulator.
- No multi-factor authentication yet; the administrator is a shared account.

Deployment

Engineers

IT leadership

IN SHORT

The app runs in Microsoft Azure as one container that both collects and serves the website, with its data on a file share that survives restarts.

Updates roll out without taking the site down. A second computer can stand by to collect if the main collector ever stops, and the app checks its own health and security every day.

6.1 Architecture

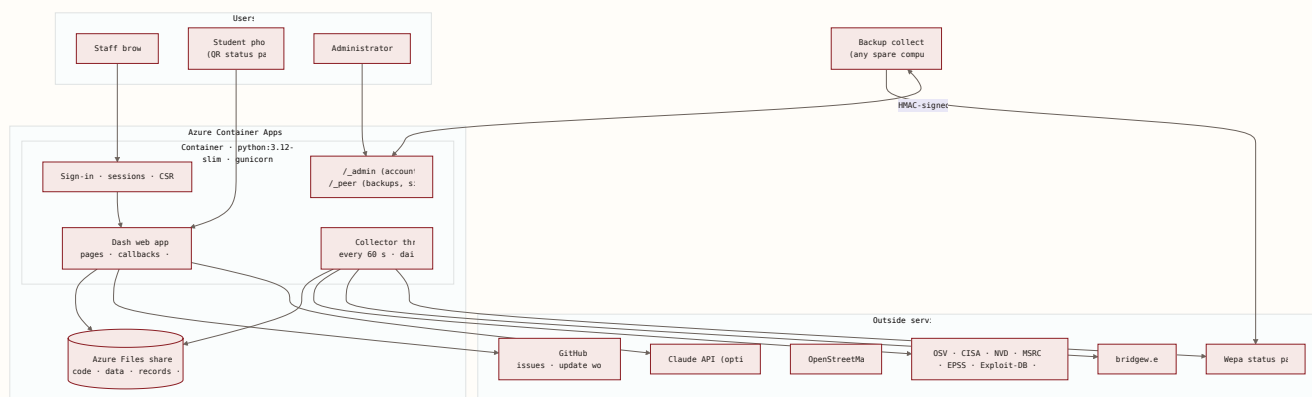


Figure 6.1 · Containers and connections. One container runs both the website and the collector; a lock on the share ensures only one copy collects. OpenStreetMap data is fetched once and kept.

6.2 Every minute

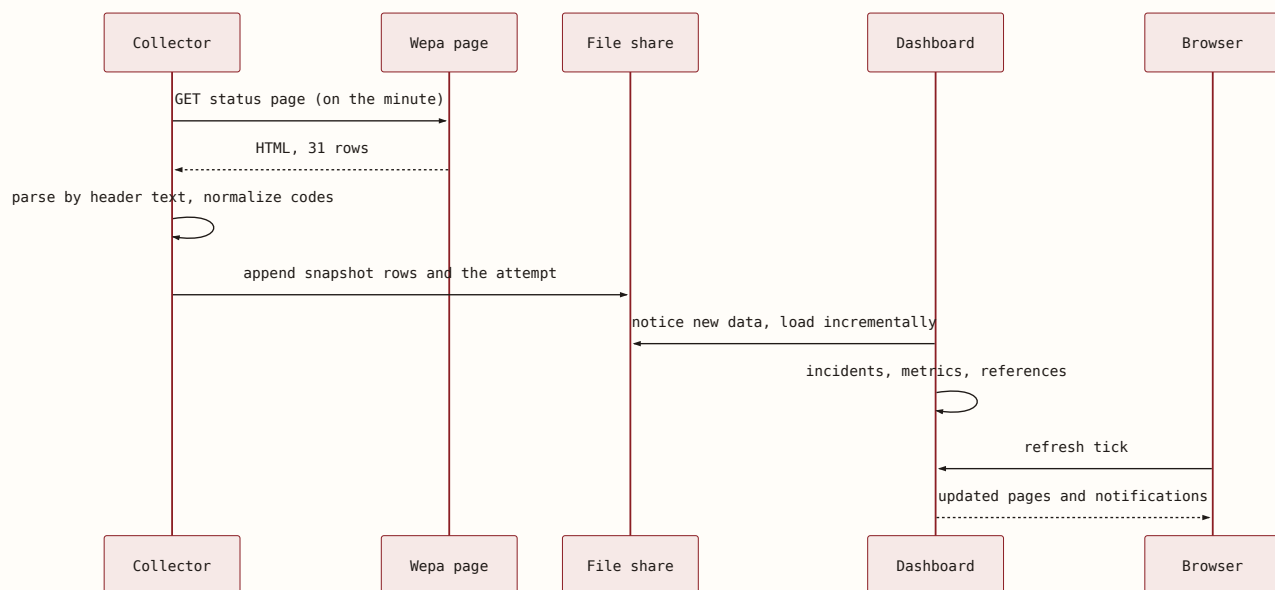


Figure 6.2 · The one-minute cycle. Once a day the collector also compacts finished days, refreshes campus data and weather, retrains the risk model, checks for vulnerabilities and sends queued feature requests.

6.3 Staying up

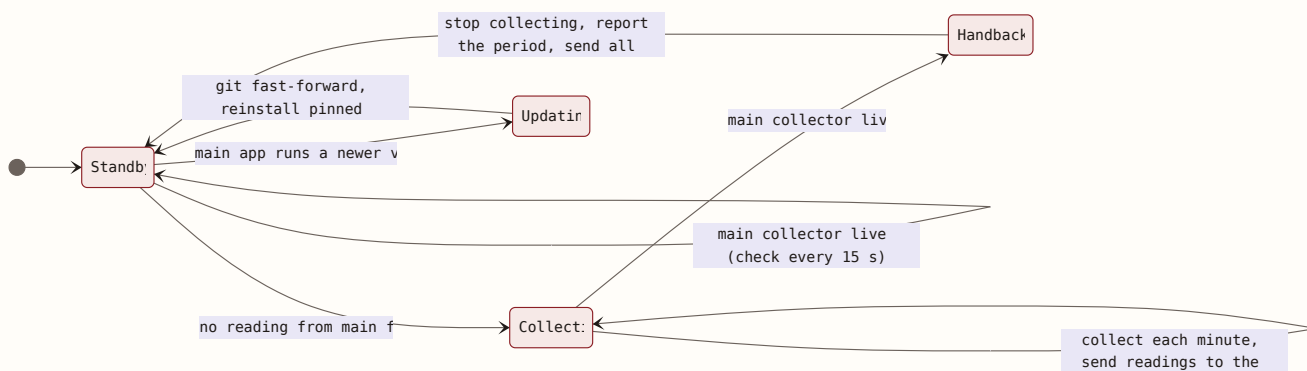


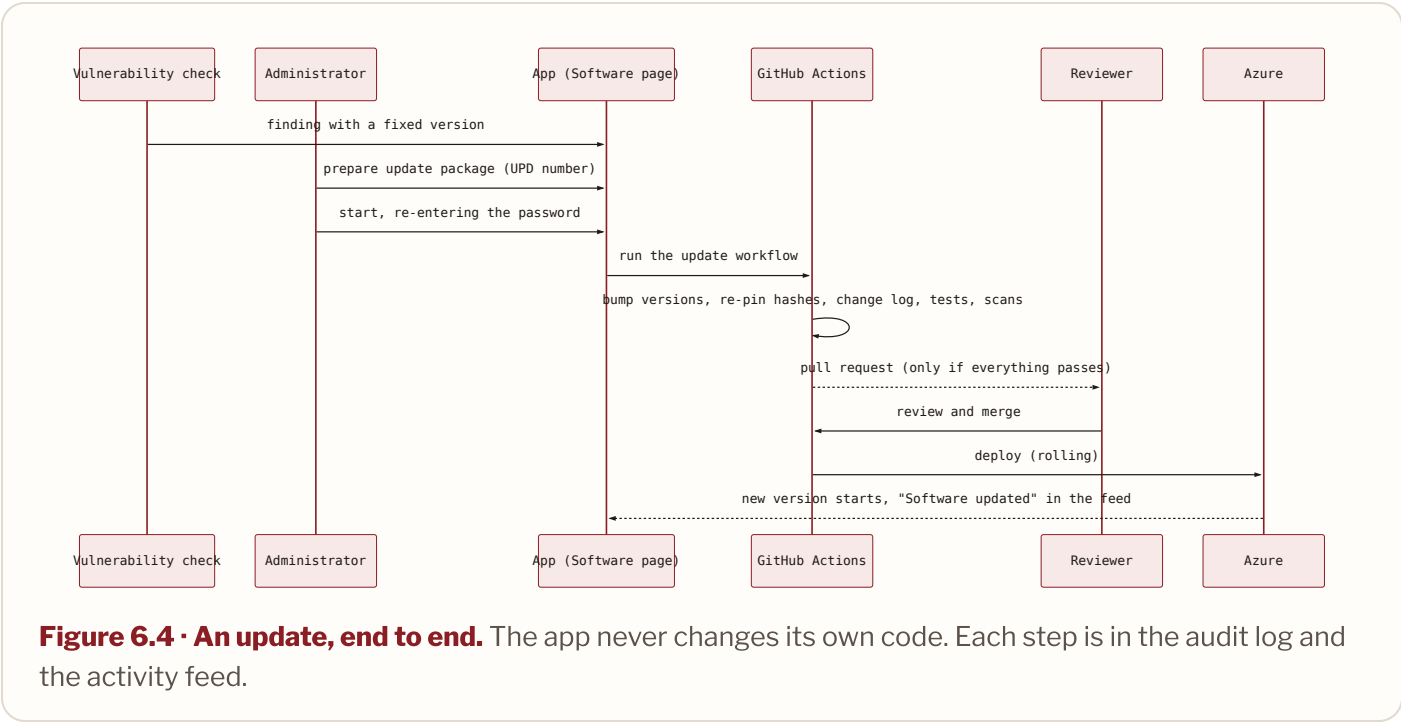
Figure 6.3 · Backup collector. Every message is signed with HMAC-SHA256 over method, path, time, a one-time number and the body; replies are signed too, so an impostor cannot make a backup stand down.

6.4 Security: confidentiality, integrity, availability

PRINCIPLE	CONTROLS
Confidentiality	Sign-in on every page; named accounts with roles; sessions end after 2 hours idle or 12 hours; NIST SP 800-63B passwords with breached-password refusal; lockouts; secrets stored as Azure secrets
Integrity	Append-only readings; hash-chained investigation history; CSRF tokens; Content Security Policy; hash-pinned package installs; signed backup messages
Availability	Rolling deploys; collector lock; backup collectors; optional second replica; gaps recorded, never guessed

The Software page carries a live OWASP Top 10 (2025) checklist with evidence for each control, and docs/SECURITY.md holds the threat model, attacker-style test results and known limits.

6.5 Updates



6.6 Monitoring and maintenance plan

WHAT	HOW OFTEN	WHERE TO LOOK	ACTION WHEN WRONG
Collector freshness, data quality	Continuous	System page, self-check	Read the live log; restart the app; check the backup
Known vulnerabilities	Daily	Software page	Prepare an update package for "Act now" first
Investigation audit trail	Every self-check	Software page	Restore records from the share's backup
Risk model gate and track record	Nightly retrain	Analytics → Forecasts and statistics	None needed; the gate hides a weak model
Reference files (stations, buildings)	When the campus changes	Stations page shows "Unassigned"	Edit <code>reference/*.csv</code> , deploy
Desk hours, holidays	Each term	<code>config.py</code> ; academic calendar	Update and deploy

How the work flows

Supervisors

IT leadership

Student staff

IN SHORT

Three standard notations describe the work around the product: BPMN for the routine outage process, CMMN for investigations (which don't follow a fixed path), and DMN for decisions (chapter 4.4). A Six Sigma view shows how the product supports continuous improvement.

7.1 Outage to resolution (BPMN-style)

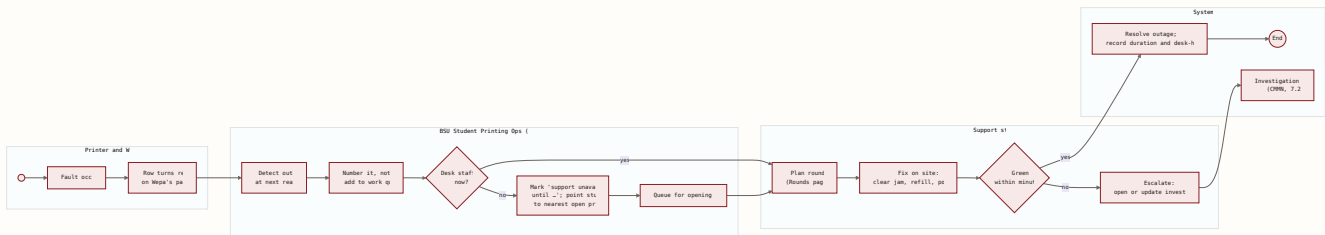


Figure 7.1 · Outage process in four lanes. Circles are start and end events, rounded boxes tasks, diamonds gateways, as in BPMN 2.0. Drawn with flowchart tooling, so shapes approximate the standard.

7.2 Investigations as a case (CMMN view)

An investigation is knowledge work: people decide what to do next. CMMN models it as a case file with stages, entry criteria (sentries) and milestones, rather than a fixed sequence.

CMMN ELEMENT	IN THE PRODUCT
Case file	The investigation record: title, station, category, impact, assignee, root cause, actions, ITSM and Wepa case numbers, linked outages, notes
Stage: Analyze	Entry sentry: an assignee is set (also sets the escalation date)
Stage: Respond	Entry sentry: a root cause, marked suspected or confirmed
Stage: Review	Entry sentry: the action taken is recorded. Exit needs a different person (separation of duties)
Milestones	Closed Complete; Closed Incomplete; Closed Cancelled (each needs a written reason)
Discretionary tasks	Add a note; link outages; download the evidence packet; override impact
Event listener	The detector opens a case when a printer has 3 or more unreachable or hardware outages in 14 days
Case file integrity	Every change is an event with the previous event's SHA-256 hash; archived 2 years after closing

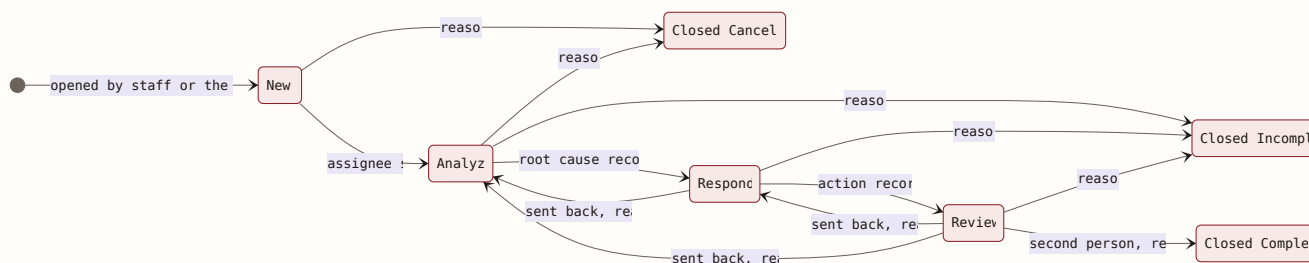


Figure 7.2 · Investigation lifecycle. Fields are editable only in New, Analyze and Respond. Time in each state, send-backs and reassignments are measured on the Investigations page.

7.3 Six Sigma: DMAIC with the product

PHASE	QUESTION	WHERE THE PRODUCT ANSWERS IT
Define	What does a defect look like to a student?	An outage: a station that cannot print. CTQs in figure 1.2
Measure	How often, how long, where, when?	Availability, repair time, time between failures, desk-hour split, data quality
Analyze	Why?	Fault ranking by cause and hour, shared outages, busy-weighted downtime, humidity test, Kaplan–Meier, investigations
Improve	What would help most?	Staffing what-ifs, printer placement, reorder points, vendor escalation with evidence
Control	Did it stay better?	c-chart with run rules, period comparisons with change tests, Executive summary

Using BSU Student Printing Ops

[Students](#)[Student staff](#)[Supervisors](#)[IT leadership](#)

IN SHORT

Sign in with the account your ResNet administrator gave you. The menu groups the pages by what you came to do (Operate, Analyze, Report, Manage). The header holds Alerts, the Assistant and your account menu; Filters and Download sit under each page's title. Staff record investigations, inventory and the key log under their own names; everything else is read-only.

8.1 For students: finding a printer that works

- 1 Scan the QR sign on the printer, where your residence hall or building has one.
- 2 The page shows whether that printer is working right now and, if not, the nearest printers that are, with the walking time. East Campus Commons counts as open to everyone; other residence-hall printers need card access.
- 3 Wepa holds print jobs online, so a job you sent can usually be released at another working station.

Student status pages are behind sign-in unless the administrator turns on public status pages (with Wepa's and BSU's agreement). They show the present only, never history.

8.2 Getting started (staff)

- 1 Open the app's address and sign in. A temporary password from the administrator must be changed the first time.
- 2 Use a password of 12 or more characters; a few unrelated words works well. Common and breached passwords are refused.

- 3 The header has three buttons: **Alerts** (stations going down or coming back, with a link to the full activity log), the **Assistant**, and your **account menu** (My account, Appearance, Directory, Suggest a feature, the administration pages, Sign out).
- 4 Under **Appearance**, choose BSU, light, dark or automatic, larger text, compact spacing, reduced motion or extra contrast. On a BSU football game day the BSU appearance becomes **Go Bears**.

8.3 Finding your way

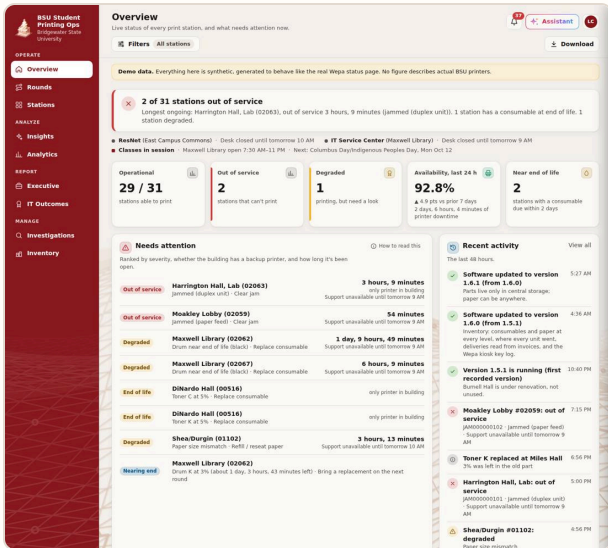
The menu is grouped by what you came to do. On a computer it runs down the left; on a phone the four most used pages sit along the bottom and **More** opens the rest.

GROUP	PAGES
Operate	Overview · Rounds · Stations
Analyze	Insights · Analytics
Report	Executive summary · IT Outcomes
Manage	Investigations · Inventory
Administration (account menu)	System health · Software and security · Activity log

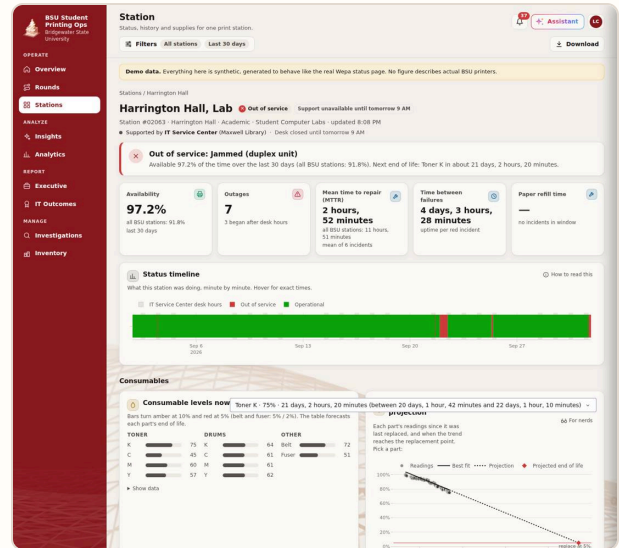
Filters sit under the page title, in the same place on every page that has them. The button shows what's applied (for example "All stations · Last 30 days"); open it to choose stations and the period, or **Clear filters**. **Download**, beside it, exports the data behind the page. Tabs along the top of a page (Analytics, Investigations, Inventory) go into the address, so a tab can be bookmarked or shared.

8.4 Pages

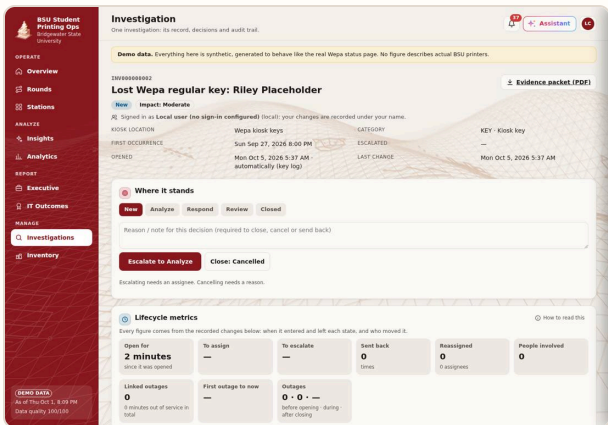
PAGE	USE IT TO	TIP
Overview	See what needs attention now: status map, work queue, end-of-life watch, recent activity	Start every shift here
Rounds	Get the fastest route to every printer that needs a visit, on foot or by van	Urgent printers come first
Stations	Search all stations; open one for its history, parts and backups	Select any chart point for its meaning
Insights	Read the story of a day, week or term; ask the data a question in plain English	Good for a supervisor's weekly note
Analytics	Reliability, faults, supplies, usage, report card, planning, forecasts and statistics	Set the period under Filters
Executive summary	Month and year to date, with what changed	Download as PDF for meetings
IT Outcomes	A print-ready feature for the IT annual report; in the launch year, a stakeholder briefing	Prints on letter paper
Investigations	Document recurring problems, missing stock and lost keys; move them through the lifecycle; download evidence for Wepa	A second person closes from Review
Inventory	Stock and its trail; record deliveries, moves, counts and write-offs; weekly paper checks; the kiosk key log; storage setup	Deliveries need a second person's approval
System health	Is the monitor healthy? Live log, data quality, visits, sign-in security, AI status	Check here if numbers look odd
Software and security	Version, self-check, vulnerabilities and their priority, OWASP checklist, updates, backups, change log	The administrator starts updates here
Activity log	Everything that happened, searchable by date	Select a version number for its change log
Assistant	Ask questions in plain English; answers cite only looked-up figures	Ctrl or Cmd + I opens it



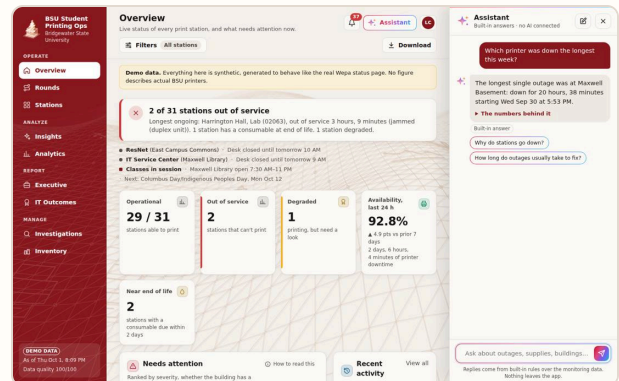
Overview. Status map, work queue and what needs attention.



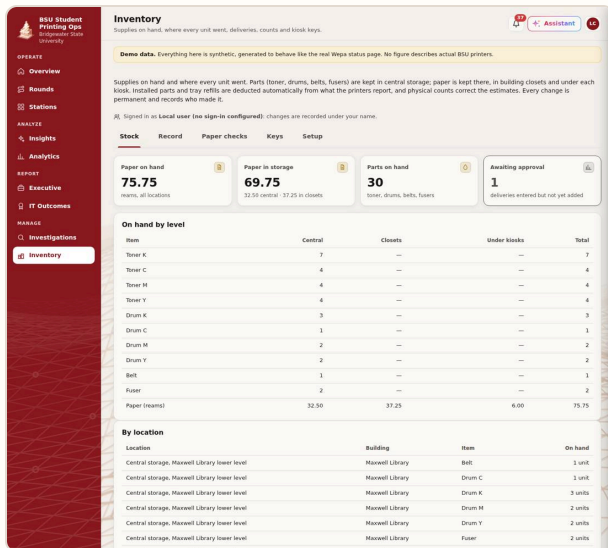
Station page. History, parts and the nearest backups.



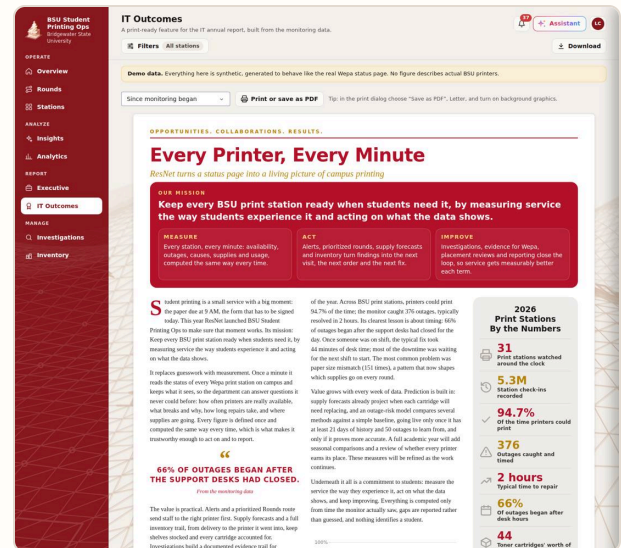
Investigation. Governed lifecycle, audit trail and evidence packet.



Assistant. Answers from read-only tools, fact-checked.



Inventory. Stock at every level and the trail of every unit.



IT Outcomes. The launch-year stakeholder briefing.

Screenshots use the synthetic demo data set.

8.5 Common tasks

RESPOND TO AN OUTAGE

- 1 On **Overview**, open the top item in the work queue, or go to **Rounds** for a route.
- 2 Fix it on site. The station turns **Operational** within a minute of Wepa's page updating.
- 3 The outage resolves itself with its reference number, duration and desk-hour split. Nothing to type.

WORK AN INVESTIGATION

- 1 Open **Investigations**. Detector-opened cases are marked; you can also open one yourself.
- 2 Assign it (moves to **Analyze**), record a root cause (**Respond**), record the action taken (**Review**).
- 3 A second person closes it. Download the evidence packet (PDF) for a Wepa case.

REPORT THE MONTH

- 1 Open **Executive**, pick the month, read the highlights.
- 2 Use **Download**, beside **Filters**, for PDF, Excel, CSV or JSON of the current filters and period.

RECEIVE A DELIVERY OF SUPPLIES

- 1 Open **Inventory** › **Record** › **Delivery received** and upload the invoice or a photo of the receipt. The lines fill in; check each against the boxes, or type them.
- 2 Submit. A second person checks it and approves it; only then is it added to stock.

COUNT PAPER (WEEKLY, BY FRIDAY)

- 1 Open **Inventory** › **Paper checks** to see what's due, then **Record** › **Counted**.
- 2 Enter reams to two decimals (a half-used ream is 0.50). If more than half a ream, or any part, is missing, an Inventory investigation opens to find out where it went.

A WEPA KEY IS LOST

- 1 Open **Inventory** › **Keys** › **Key returned, lost or found**, choose the key and **Lost**, and say what happened.
- 2 A Kiosk key investigation opens. It requires a counseling conversation with management, recorded on the investigation; for a master key, decide with Wepa whether the locks must change.

SUGGEST A FEATURE OR REPORT A PROBLEM

- 1 In your account menu, choose **Suggest a feature**.
- 2 Pick the kind, give a title and details, and send. It gets a REQ number and becomes a GitHub issue; your name is not included unless the administrator turned that on.

8.6 For the administrator

TASK	COMMAND OR PLACE
Add a staff account	<code>python -m wepa_monitor users add jsmith --name "Jordan Smith" --email jsmith@bridg</code>
Reset, deactivate, list	<code>users reset jsmith</code> · <code>users deactivate jsmith</code> · <code>users list</code>
Deploy or update	<code>./deploy/azure-containerapps/deploy.sh</code> (from a laptop or Azure Cloud Shell)
Add keys safely	<code>ASK_SECRETS=1 ./deploy/azure-containerapps/deploy.sh</code> (hidden prompts)
Check the app	<code>python -m wepa_monitor selfcheck --vulns</code> , or the Software page
Start a backup collector	<code>python -m wepa_monitor backup --primary https://<app></code> with <code>WEPA_PEER_KEY</code> set
Start a security update	Software page → Prepare update package → Start update (password again)

8.7 Frequently asked questions

Why does a station say "No signal"?

It hasn't appeared on Wepa's page for 5 minutes. Its status is unknown, and that time is left out of every rate.

Why "support unavailable until ..."?

The team that supports that printer has no desk open right now; the time is when it next opens.

Why does a figure say "insufficient data"?

Too few incidents or too little observed time to be reliable. It appears once there is enough evidence.

Why isn't the outage-risk model shown?

It hasn't beaten the simple baseline on held-out weeks yet; the page says which condition is not met.

Is "In progress" ever shown?

No. The monitor can see that a problem is open, not whether someone is working on it, so it says Ongoing.

Standards and recommendations

IT leadership

Engineers

Data scientists

IN SHORT

The product was built with recognized standards and professional codes in mind. This chapter maps each one to what the product does, says honestly where it falls short, and recommends next steps. It is a self-assessment: no certification or formal audit is claimed.

9.1 Alignment matrix

STANDARD OR CODE	WHAT IT ASKS, IN BRIEF	HOW THE PRODUCT ALIGNS	STATUS
CRISP-DM	Six phases from business understanding to deployment, iterated	This document's structure; nightly retraining and self-checks keep the cycle running	Aligned
ISO/IEC 25010 (product quality)	Quality characteristics such as reliability, security, maintainability, usability	Quality requirements NFR-01 to NFR-12, each with a mechanism	Aligned
ISO/IEC/IEEE 26514 (information for users)	Audience analysis, task-oriented content, consistent terminology	Reading paths by audience; task steps; one status vocabulary; glossary	Aligned
ISO/IEC/IEEE 15289 (life-cycle information items)	Defined content for plans, descriptions and records	Document control, requirements, architecture, change log, test evidence	Partial
ISO 8601	Unambiguous dates and times	Stored in UTC ISO 8601; shown in local time with the time zone named	Aligned
ISO/IEC 27001 (information security)	Risk-based security controls and management	Access control, cryptography, logging, secure development, supplier checks; no formal ISMS	Partial
OWASP Top 10 (2025)	The ten most critical web application risks	Live checklist with evidence in the app: 7 met, 3 partial on a correctly configured deployment	Partial
NIST SP 800-63B	Password and session guidance	Length over complexity, breached-password check, no forced expiry, lockouts; MFA not yet	Partial
BPMN 2.0, CMMN 1.1, DMN 1.x (OMG)	Notations for processes, cases and decisions	Figures 7.1 and 7.2 and the decision tables in 4.4	Aligned
Six Sigma (DMAIC, SIPOC, SPC)	Define and measure defects; control with statistics	SIPOC, CTQ tree, DMAIC map, c-chart with run rules	Aligned

ACM Code of Ethics (2018)	Contribute to society, avoid harm, be honest, respect privacy, keep systems secure	Mapped in 9.2	Aligned
ASA Ethical Guidelines for Statistical Practice	Integrity of data and methods; responsibilities to stakeholders and data subjects	Mapped in 9.3	Aligned
DASCA role families	Data engineering, analysis and science as distinct competencies	Mapped in 9.4	Aligned
Data ecology	Data as an ecosystem of sources, stewards, flows and uses	Mapped in 9.5	Aligned
WCAG 2.2	Accessible web content	Keyboard focus, contrast option, reduced motion, text sizes; not formally audited	Partial

9.2 ACM Code of Ethics

PRINCIPLE	IN PRACTICE
1.1 Contribute to society and human well-being	The product's goal is that students can print when they need to
1.2 Avoid harm	Data is about printers, never people; the AI never guesses who caused a problem
1.3 Be honest and trustworthy	No imputed data; evidence gates; demo data labelled on every page
1.6 Respect privacy	Visits shortened to the network; no passwords logged; names kept out of public issues
2.5 Give comprehensive evaluations, including risks	Chapter 5; known limits stated in the app and here
2.9 Design and implement systems that are robustly and usably secure	Chapter 6.4; docs/SECURITY.md; attacker-style tests
3.7 Recognize and take special care of systems that become integrated into society	Vendor terms respected; public pages opt-in only

9.3 ASA Ethical Guidelines for Statistical Practice

GUIDELINE AREA	IN PRACTICE
Professional integrity and accountability	Methods named and documented; sample sizes on every statistic; uncertainty shown as intervals
Integrity of data and methods	Raw data never edited; censoring handled (Kaplan–Meier); confounding controlled (Mantel–Haenszel); models gated against a baseline
Responsibilities to stakeholders	Plain-language summaries; "not enough evidence" said plainly; correlation not presented as cause
Responsibilities to data subjects	Minimal personal data; staff directory visibility limited by role

9.4 DASCA role families

ROLE FAMILY	WORK DEMONSTRATED IN THIS PRODUCT
Data engineering	Resilient once-a-minute ingestion; append-only storage with Parquet compaction; incremental loading; imports and de-duplication; backup collection with signed transfer
Data analysis	KPI definitions with gates; dashboards; stories; exports; executive reporting
Data science	Survival analysis, SPC, robust regression, empirical Bayes, Monte Carlo, stratified rate ratios, gated machine learning with walk-forward evaluation, an evidence-bound AI analyst

9.5 Data ecology

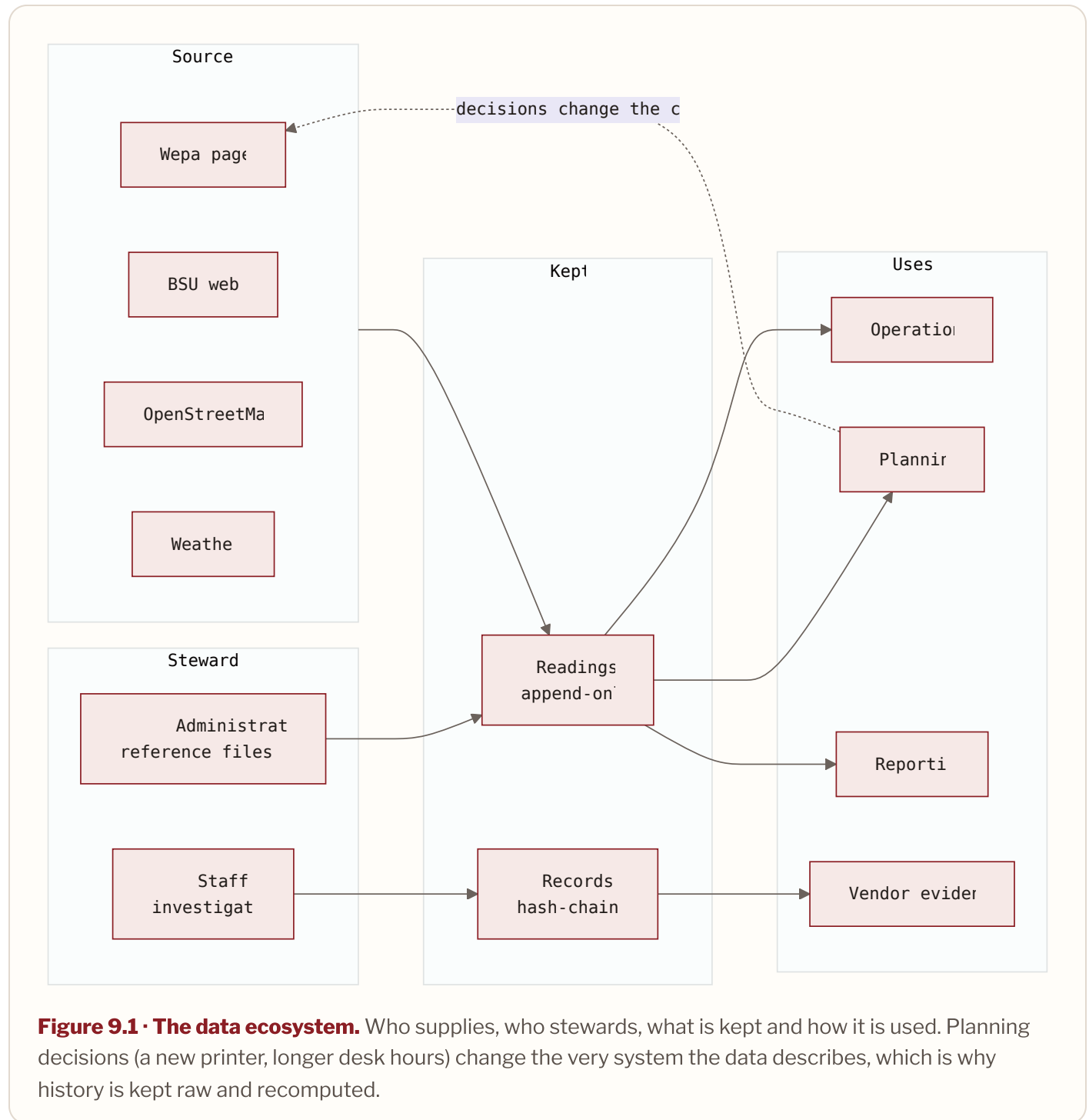


Figure 9.1 • The data ecosystem. Who supplies, who stewards, what is kept and how it is used. Planning decisions (a new printer, longer desk hours) change the very system the data describes, which is why history is kept raw and recomputed.

9.6 Documentation and standards recommendations

#	RECOMMENDATION	WHY	EFFORT
1	Single sign-on through BSU's Microsoft Entra ID with MFA	Closes the largest security gap; ties actions to people	Medium, needs BSU IT
2	Formal WCAG 2.2 AA audit with assistive technology	Accessibility claims should be tested, not assumed	Low
3	Architecture decision records (docs/ad1/)	Keeps the reasons behind choices (no imputation, gated ML) with the code	Low
4	Software bill of materials (CycloneDX) per release	Supply-chain transparency; complements the vulnerability check	Low
5	Data retention policy for readings, visits and records	States how long each kind of data is kept and why	Low, needs a decision owner
6	Model card for the outage-risk model, published with each retrain	Documents data, metrics, limits and intended use (the app already computes these)	Low
7	ITSM integration by reference number (read-only link to tickets)	Traceability without duplicating BSU's ticketing system	Medium
8	Run the container as an unprivileged user	Limits the damage of any future flaw	Low
9	Written agreement with Wepa and BSU before any public use of status data	Required by Wepa's terms	Low
10	Docs as code: build this document in CI on each release	Documentation stays in step with the product	Low

Reference

A.1 Configuration

SETTING	PURPOSE
WEPA_DATA_DIR , WEPA_CAMPUS_DIR	Where readings and campus files are kept
WEPA_COLLECT	1 to collect in this process, 0 for dashboard only
WEPA_BASIC_AUTH , WEPA_SECRET_KEY	Administrator credential (hashed) and session signing key
WEPA_BREACH_CHECK	0 turns off the breached-password lookup
WEPA_PUBLIC_STATUS	1 makes student status pages public (agreement required)
WEPA_AI_PROVIDER , WEPA_AI_MODEL , ANTHROPIC_API_KEY , WEPA_AI_MAX_PER_HOUR	Optional AI analyst and its spending cap
WEPA_VULN_CHECK , WEPA_THREAT_INTEL , WEPA_NVD_API_KEY	Vulnerability checks and threat-intelligence look-ups
WEPA_GITHUB_TOKEN , WEPA_GITHUB_ISSUES_TOKEN , WEPA_GITHUB_REPO	Starting updates; filing feature requests
WEPA_FEEDBACK_NAMES	1 adds the sender's name to feature-request issues
WEPA_PEER_KEY	Shared key for backup collectors (32+ characters)
WEPA_SUPPLY_LEAD_DAYS	Days a supply order takes (default 7)

A.2 Commands

COMMAND	DOES
<code>python -m wepa_monitor start</code>	Collect and run the dashboard on one computer
<code>python -m wepa_monitor demo --days 120</code>	Generate synthetic demo history
<code>python -m wepa_monitor dashboard --demo</code>	Run the dashboard on demo data
<code>python -m wepa_monitor campus --force</code>	Refresh calendar, halls, library hours and classes
<code>python -m wepa_monitor train-risk</code>	Train and evaluate the outage-risk model now
<code>python -m wepa_monitor users ...</code>	Manage staff accounts on the running app
<code>python -m wepa_monitor selfcheck [--vulns]</code>	Run the self-check
<code>python -m wepa_monitor backup --primary URL</code>	Stand by as a backup collector
<code>python -m wepa_monitor export --tag mac</code>	Package local data for import elsewhere
<code>python -m wepa_monitor kml</code>	Building pins for Google Earth

A.3 Release history

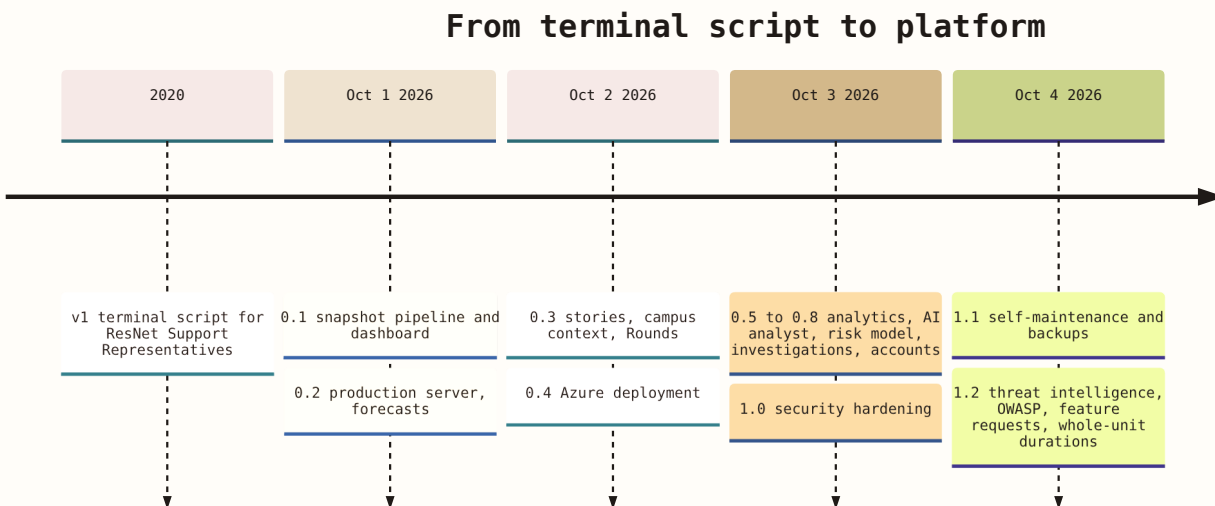


Figure A.1 · Releases. From `CHANGELOG.md`; each release links to its source commits in the app's Change log page.

A.4 Glossary

Availability

Share of observed time a printer could print.

Censored

An incident whose start or end is outside what was observed; kept, but handled so it doesn't bias averages.

CRISP-DM

Cross-Industry Standard Process for Data Mining: business understanding, data understanding, data preparation, modeling, evaluation, deployment.

CTQ

Critical to quality: a measurable characteristic that matters to the customer.

CVE, GHSA

Public identifiers for a security vulnerability (Common Vulnerabilities and Exposures; GitHub Security Advisory).

Degraded

Printing, with a warning.

Desk hours

When a support team's desk is staffed.

EPSS

Exploit Prediction Scoring System: probability a CVE is exploited in the next 30 days.

KEV

CISA's Known Exploited Vulnerabilities catalog.

MTBF, MTTR	Mean time between failures; mean time to repair.
No signal	No reading for 5 minutes; status unknown.
Observed time	Time covered by readings, at most 5 minutes per reading.
Out of service	Cannot print (a red row on Wepa's page).
ResNet	BSU's residential network and technology support team, which supports the residence-hall printers.
RSR	ResNet Support Representative: a student staff member.
SIPOC	Suppliers, inputs, process, outputs, customers.

A.5 Sources and further reading

- Chapman, P. et al. (2000). *CRISP-DM 1.0: Step-by-step data mining guide*.
- ISO/IEC 25010, ISO/IEC/IEEE 26514, ISO/IEC/IEEE 15289, ISO 8601, ISO/IEC 27001.
- Object Management Group: BPMN 2.0, CMMN 1.1, DMN.
- OWASP Top 10:2025, owasp.org/Top10/2025.
- NIST Special Publication 800-63B, Digital Identity Guidelines.
- ACM Code of Ethics and Professional Conduct (2018).
- American Statistical Association, Ethical Guidelines for Statistical Practice.
- Data Science Council of America (DASCA), credential frameworks.
- Kaplan, E. L. and Meier, P. (1958); Theil, H. (1950) and Sen, P. K. (1968); Mantel, N. and Haenszel, W. (1959).
- OpenStreetMap contributors (ODbL); Open-Meteo (CC BY 4.0); OSV.dev; CISA KEV; NIST NVD; Microsoft MSRC; FIRST EPSS; Exploit-DB; Metasploit Framework.

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Designed, built and documented by **Nicholas Freberg**, Bridgewater State University Class of 2020 and former BSU ResNet Support Representative (2018–2020), for the students and staff who keep campus printing.

Set in Libre Franklin, Source Serif 4 and IBM Plex Mono, in BSU crimson on campus stone. An independent portfolio project; not an official publication of Bridgewater State University or Wepa. Product names belong to their owners.